

Adaptive Distributional Security

A Framework for Input-Adaptive Cryptography*

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Abstract

It is often desirable to break cryptographic primitives into two components: an input-independent *offline* component, and a cheap *online* component used when inputs arrive. Security of such online/offline primitives must be proved in the *input-adaptive* setting: the adversary chooses its input adaptively, based on what it sees in the offline-phase. Proving security in the input-adaptive setting can be difficult, particularly when one wishes to achieve simulation security and avoid idealized objects like a random oracle (RO).

This work proposes a framework for reasoning about input-adaptive primitives: *adaptive distributional security* (ADS). Roughly, an ADS primitive provides security when it is used with inputs drawn from one of two distributions that are themselves hard to distinguish. ADS is useful as a framework for the following reasons:

- An ADS definition can often circumvent impossibility results imposed on the corresponding simulation-based definition. This allows us to decrease the online-cost of primitives, albeit by using a weaker notion of security.
- With care, one can typically upgrade an ADS-secure object into a simulation-secure object (by increasing cost in the online-phase).
- ADS is robust, in the sense that (1) it enables a form of composition and (2) interesting ADS primitives are highly interconnected in terms of which objects imply which other objects.
- Many useful ADS-secure objects are plausibly secure from straightforward symmetric-key cryptography.

We start by defining the notion of an ADS encryption (ADE) scheme. A notion of input-adaptive encryption can be easily achieved from the ROs, and the ADE definition can be understood as capturing the concrete property provided by ROs that is sufficient to achieve input-adaptivity. From there, we use ADE to achieve ADS variants of garbled circuits and oblivious transfer, to achieve simulation-secure garbled circuits, oblivious transfer, and two-party computation, and prove interconnectedness of these primitives. In sum, this results in a family of objects with extremely cheap online-cost.

Keywords: Input-Adaptivity; Simulators; MPC; Garbled Circuits

*This work is inspired by a question asked by Prof. Adam Groce at Reed College about whether simulation-based security is too strong in some settings, leading to artifacts in protocol design [A. Groce, Personal Communication, Spring 2020].

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1 Introduction

The Input-Adaptive Setting. Many cryptographic primitives/protocols can be made more flexible by splitting execution into two phases: an *offline-phase*, which runs before party inputs are known, and an *online-phase*. Often, it is possible to move most protocol cost to the offline-phase, resulting in a highly efficient online-phase. This can be desirable, as it allows the parties to complete most of the work when their computational resources are free. Once the input becomes available, which might happen when computational resources are in high demand, they complete the protocol in a timely manner.

While it can be desirable to split a protocol up this way, proving the split secure can be problematic. The challenge is that the adversary can base its protocol input on the offline-phase’s protocol transcript. We refer to this general setting as the *input-adaptive* setting: the adversary adaptively chooses its input, based on the offline-phase. This setting presents a challenge to achieving provable security as, for example, there is no way to force the adversary to commit to its input before the offline-phase begins—that input is simply not yet defined. Because of this, it can be challenging to show that input-adaptivity does not give the adversary noticeable advantage. This challenge is especially apparent when trying to prove an offline/online primitive satisfies strong security properties, such as simulation-based security.

Proving that a primitive achieves a notion of simulation security is highly desirable, as it is well known that such notions are composable, allowing to quickly and modularly construct sophisticated protocols. But in the input-adaptive setting it can be challenging to build a simulator, as the simulator must emulate adversarial behavior, and, very roughly, it is hard to perform this emulation as the simulator must commit to an offline-phase transcript that will (from the adversary’s point of view) correspond to an input that has not yet been chosen.

An Example Primitive: Garbling Schemes. Consider the well-studied problem of adaptively secure garbled circuit (GC) schemes [BHR12b]. GC is a powerful primitive allowing a garbler to construct from a circuit C a privacy-preserving encoding $\tilde{C}$ [Yao86]. Given $\tilde{C}$ and an encoded input $\tilde{x}$, an evaluator can obtain $C(x)$ while learning nothing beyond the output. Specifically, the cleartext input x is kept private. GC naturally enables round-efficient two-party computation (2PC) protocols.

To improve online efficiency, one might consider sending $\tilde{C}$ in advance, before the input x is known. Then, in the online-phase, the parties seem to only need to exchange an input encoding $\tilde{x}$. This proposed construction could reduce online communication such that it is independent of the circuit size, and even of the circuit’s output size, a clear instance of the advantage of input-adaptivity.

Unfortunately, building a garbling scheme that is secure in the input-adaptive setting is notoriously challenging. In fact, if we consider the above attempt—where online communication is independent of the output size—proving simulation security is not just challenging, it is *impossible*. Specifically, any simulation-secure adaptive garbling scheme must have communication that scales with the size of the output [AIKW13]. Notably, this kind of impossibility result—where online-phase cost must scale with output size—is somewhat ubiquitous in the input-adaptive setting. For instance, a similar restriction also applies to an input-adaptive variant of simulation-secure 2PC [HW15].

Returning to our example of adaptive garbling, we note that even when one adjusts their garbling scheme to avoid this impossibility result, achieving input-adaptive security remains a challenge.

Input-Adaptivity in Idealized Models. Interestingly, the challenge of input-adaptivity can often be essentially eliminated by considering security in an idealized model, such as the random oracle (RO) model (ROM). Roughly, when working in the ROM, we can adjust the offline-phase such that particular messages are encrypted with the output of an RO. This adjustment allows simulation security, as the simulator can program the RO to adjust the offline-phase transcript after the adversary chooses its input. This strategy is in particular known to work for garbling schemes [BHR12a].

The strategy of encrypting with ROs is troubling, in particular because it can circumvent the kinds of impossibility results discussed above. Note that if one limits the flexibility of the ROM by preventing the simulator from programming the RO (known as the non-programmable ROM (NPROM) [FLR⁺10]),

impossibility results once again apply [AIKW13, BHKO23]. However, even when restricting to the NPROM, there still seems to be a very significant gap between what we can achieve from the ROM and what we can achieve from standard model assumptions. As an example, the standard model adaptive garbling schemes rely either on sophisticated and asymptotically expensive (in the online-phase) tricks [HJO⁺16], or on public-key cryptographic assumptions [GOS18, GS18], which are not needed in the non-adaptive (or *selective*) setting; in contrast, many state-of-the-art garbling schemes that were designed for the selective setting are *automatically* secure in the adaptive setting when the scheme’s underlying hash function is treated as an NPRO [BHKO23, GYW⁺23].

Our goals in this work. In this work, we are interested in studying the theoretical foundations of input-adaptive security. Specifically, we explore and propose security definitions that admit simple adaptively-secure constructions.

More concretely and based on our above discussion, we address two questions:

- Simulation-secure input-adaptive primitives generally require an online-phase that scales with output size. *When can we achieve input-adaptive primitives whose online-cost does not scale with the output size?*
- There is often a significant efficiency gap between standard-model-based input-adaptive primitives and corresponding primitives built in the ROM, even when we restrict use of the RO, e.g. via the NPROM. *What concrete property does RO provide that makes achieving input-adaptive security so much easier?*

1.1 Our Contribution

In seeking to answer the above questions, we propose a security framework that captures much of the nuance of input-adaptivity. We refer to this framework as the *adaptive distributional (AD) security (ADS)* framework.

ADS vs Simulation-Security. ADS-based security definitions tend to avoid impossibility results associated with simulation security. For instance, our notion of adaptive-distributional GC (ADGC) circumvents the need for the online-phase to scale with output size. This is because the notion of ADS is weaker than that of simulation-based security.

Despite this relative weakness, we show examples (Theorems 14 and 15) where an ADS-based object can be used as a black-box to construct an analogous object that attains adaptive simulation-based security and (within constant factors of) optimal online-cost. Thus, ADS as a framework allows one to plug together protocols, each of which do not scale in the online-phase with the size of their output, then to upgrade the result with simulation security by paying for the output size of the outermost protocol only.

In other words, the ADS framework *is not* meant as an *alternative* to simulation security, but rather as a framework for defining primitives *for the purpose of achieving* efficient adaptive simulation-based security.

Encryption Definition for Input-Adaptivity. As discussed above, many input-adaptive problems become far easier in the ROM. One attractive property of such RO-based solutions is that they achieve a balance between security and practicality, as we can give fast candidate constructions to instantiate the RO (such as SHA or, in certain instances, AES). However, it is well known that the ROM—programmable or not—is uninstantiable [CGH98], which calls into question the security of such schemes.

It would be far more desirable to find a falsifiable complexity assumption sufficient to prove security of particular input-adaptive solutions. Such an assumption might be plausibly instantiated in the same way as RO, but would allow for far greater confidence that the resulting system may indeed be secure. Returning to our example of GC, an analogous approach was taken with respect to the popular Free XOR optimization [KS08]. While standard garbling can be proven secure given only a one-way function [LP09], the Free XOR optimization originally leveraged RO. Follow-on work defined a concrete property of a hash function called *circular correlation robustness* (CCR) that is sufficient to prove security of the Free XOR

technique [CKKZ12]. This CCR definition is trivially achieved by ROs, so there is no change in efficiency, and its introduction increases confidence that Free XOR is plausibly secure.

As one of our main contributions, we use our ADS framework to specify a security definition for encryption called adaptive distributional encryption (ADE). Analogous to CCR and Free XOR, ADE provides a falsifiable complexity assumption¹ on an encryption scheme that we demonstrate is sufficient to achieve input-adaptivity for several problems. Even the strongest variant of ADE is *trivially* achieved by RO. Thus, ADE provides insight into what sorts of security properties are sufficient to immediately achieve input-adaptivity, in a similar way to the ROM.

Additional ADS Constructions. Beyond ADE, we also apply the ADS framework to a variant² of oblivious transfer (OT) and garbled circuits, called adaptive distributional OT (ADOT) and adaptive distributional GC (ADGC). For each of these primitives, we are able to obtain AD security with a succinct online-phase. In particular, the online-cost of the OT and GC variants is the size of GC labels (for any selectively secure scheme), and the cost of ADE applied to garbled material.

To top off the contributions of the new framework, we also make use of recent advances in GC security from the decisional composite residuosity (DCR) assumption [Pai99] to achieve a very succinct and highly practical ADOT scheme that has an online-phase of $O(n + \text{poly}(\lambda))$, along with similar implications for ADGC and ADE.

We show that this new ADOT scheme can be used to achieve near-optimal online-cost, constant online round 2PC from ADE and DCR with online-cost $2m + O(n + \text{poly}(\lambda))$, where n is the number of input bits and m the number of output bits.³ We note that prior work [AIKW13, DLL24] was able to achieve similar results, but required unreserved use of the ROM, which we avoid.

1.2 Overview of Our Approach

To demonstrate the ADS framework, we describe our first concrete contribution: a security definition for an encryption scheme that captures the nuance of input-adaptivity, which we call *adaptive distributional encryption* (ADE).

In its simplest form, an ADE scheme has the same form as any other encryption scheme: one can encrypt a message to get a ciphertext and a short key, and then later use the ciphertext and key to get the message back. ADE is designed to be “secure” in the following specific setting:

Suppose we give an adversary an encryption to a message in the offline-phase. Then the adversary does some other side operation in the offline-phase, perhaps including running other cryptography related to the encrypted message. Lastly, we give the adversary the secret key so that it might learn the message in a succinct online-phase.

We might wish that the encryption scheme maintains security as if the message has simply not been sent until the online-phase. Similar attempts at such security notions have been proposed in the past, such as with non-committing encryption [Nie02]. Non-committing encryption has a simulation-based security definition in which a simulator must emit a ciphertext, and then later is given some message and is tasked with producing a decryption key that decrypts the unrelated ciphertext to the given message. Such a scheme clearly satisfies the criteria we posited above, but the same work of [Nie02] shows that succinct-key non-committing encryption cannot be achieved in the standard model, despite being straightforward in the ROM.

ADE can be understood as a weakening of non-committing encryption that is plausibly instantiable and is strong enough to construct many adaptive protocols with attractive online-phase communication costs.

¹ADE is not what is called a “first degree” falsifiable assumption. More about this at the end of Section 1.5.

²In our considered variant of OT, the sender’s messages are fixed in the offline-phase, and the receiver’s choice bits are learned in the online-phase. This primitive is, for example, useful as a building block in 2PC protocols, where the sender’s messages might be cryptographic labels needed to evaluate a garbled circuit.

³The $2m$ factor comes from the fact that only one party gets the output and needs to send it to the other party, but in the case that only one party needs to get input this can be avoided.

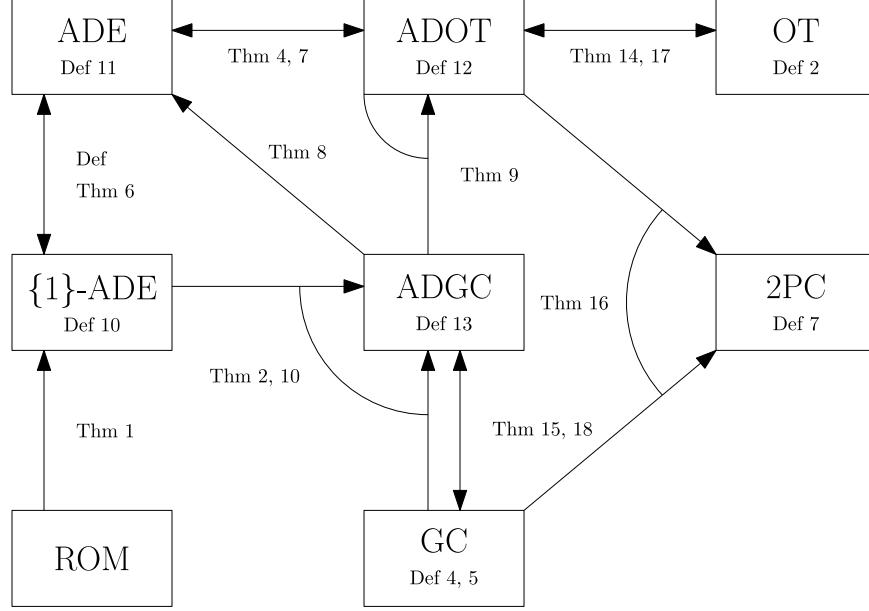


Figure 1: An implication diagram illustrating how the primitives in this work relate to each other and the reference definitions and theorems. Arrows $A \rightarrow B$ means that primitive A can be used to construct B , maintaining succinctness when possible. Arrows/primitives connected by arcs are implications that require multiple primitives. For example, ADGC can be built from $\{1\}$ -ADE *and* GC as seen in Theorems 2 and 10.

In order to define security of an ADE scheme, we first require an intermediate definition that relates two “similar” distributions, which we refer to as *distributional indistinguishability*.⁴ Here, we consider a distribution D_0 (resp. D_1) that outputs a message $\bar{m}_0$ (resp. $\bar{m}_1$), as well as a “context” σ_0, ξ_0 (resp. σ_1, ξ_1). The notion of a context is central to ADS. In short, the side-information σ corresponds to the notion of “what the adversary sees in the offline-phase”, and ξ is an efficient function that corresponds to the notion of “what the adversary sees in the online-phase”. Distributional indistinguishability is defined with respect to the following game (see Definition 8 for formal presentation):

- First, the challenger uniformly chooses either D_0 or D_1 , and samples $\sigma, \xi, \bar{m}$ from the chosen distribution.
- Next, the challenger gives the adversary the offline-phase information σ . In response, the adversary is allowed to submit a “query” to the online-phase, called η .
- Finally, the challenger provides the adversary with online-phase information $\xi(\eta)$ and the message $\bar{m}$.

Distributions D_0 and D_1 are indistinguishable if no adversary can reliably guess whether the challenger chose D_0 or D_1 .

Now we are ready to define security of an ADE scheme. In particular, the security of an ADE scheme is defined with respect to an arbitrary pair of indistinguishable distributions D_0 and D_1 (see Definition 10 for formal presentation):

- First the challenger uniformly chooses either D_0 or D_1 , and samples $\sigma, \xi, \bar{m}$ from the chosen distribution.
- Next, the challenger uses the encryption scheme to encrypt $\bar{m}$, resulting in ciphertext c and master key k .

⁴The property is actually called $\mathcal{X}_n$ -distributional indistinguishability, with this instance referring to $\{1\}$ -distributional indistinguishability. We simplify the notation here for clarity of exposition.

- The challenger gives the adversary both σ and c . In response, the adversary is allowed to submit a “query” to the online-phase, called η .
- The challenger gives k and $\xi(\eta)$ to the adversary.

The encryption scheme satisfies ADE security if no adversary can reliably guess whether the challenger chose D_0 or D_1 . Informally, ADE captures the notion that if two processes are indistinguishable, given that the adversary does not have access to some message in the offline-phase, then it should still be indistinguishable when only an encrypted version of that message is given in the offline-phase. Importantly however, the distributional nature of the drawn $\sigma, \xi, \overline{m}$ enforce a sort of “plausibility” regarding the encrypted elements, unlike non-committing encryption, which could be forced to decrypt the message to some value computed using side information, even though this could not have been known during the encryption step and could even depend on adversarial decisions. This allows our construction of ADE from ROM (Theorem 1) to avoid using the RO techniques that make non-committing encryption uninstantiable, and so ADE is plausibly instantiable when non-committing encryption is not.⁵

We emphasize the similar structure of our above two ADS-style definitions. Each of our security definitions will follow the same structure, where there is a notion of an offline- and online-phase, as well as contexts σ and ξ . It is in this sense that ADS is a *framework* for security.

The compositional nature of ADS is also readily apparent. Notice that we can reformulate the definition of ADE security as an *instance* of distribution indistinguishability, albeit where the message is trivial (i.e., empty). In particular, in the ADE scheme offline-phase, the challenger sends both side information σ and ciphertext c , but we can imagine absorbing the ciphertext c into a larger piece of side information $\sigma' = (\sigma, c)$; similarly, we can imagine absorbing the opening key into the online-phase information $\xi'(\eta) = (\xi(\eta), k)$. The resulting σ', ξ' can be packaged into a distribution of the same form as the above notion of distribution indistinguishability. More generally, when constructing an ADS object that delegates a task to some other ADS object in a compositional manner, we can absorb the callee’s side information σ, ξ into that of the caller.

We remark that the ADE security definition is somewhat similar to that of so-called SOA-CPA encryption [BHY09, DNRS99, FHKW10, LDL⁺14, ORV14], except that we are crucially interested in also having a succinct opening functionality, and our notion of security is slightly stronger in some contexts.⁶

ADE exemplifies the dynamic of ADS, and in this work we explore applying this framework to other primitives, including oblivious transfer (Definition 12) and garbling schemes (Definition 13).

1.3 Adaptive Security Without Modification; Multiplier-1 ADE

One interesting goal is to prove that existing, off-the-shelf cryptographic techniques—which were proven secure in the selective setting—are also adaptively secure without modification, when some assumption is made about the underlying encryption mechanism. In particular, we examine the unmodified adaptive security of garbling schemes. While many unmodified garbling schemes have been proved adaptively secure in the random oracle model [BHKO23, GYW⁺23], it remains open whether there could exist a plausibly-instantiable standard model assumption that can show adaptive security of unmodified state-of-the-art schemes for all poly-size circuits [KKPW21].⁷

Our GC constructions from ADE *do require minor modifications* to existing protocols to achieve adaptive security. We find that applying ADE to existing schemes by directly replacing the encryption primitive or hash function used in the scheme does not immediately work; however, we give a slight strengthening to ADE, called multiplier-1 ADE (M1-ADE) that *is* able to achieve adaptive security of the garbling scheme of [AIK11]⁸ without modification.

⁵For more information on ROM techniques and uninstantiability results see [BHKO23, FLR⁺10].

⁶Only projective ADE schemes are comparable to distributional versions of symmetric-key SOA security.

⁷Recall that the scheme of [JW16] does get adaptive security for unmodified Yao’s scheme for log-depth circuits.

⁸We find that the GC scheme of [AIK11] is a good stand-in for Yao-style garbling schemes in general. The techniques presented should also work for Yao’s scheme [Yao86].

M1-ADE is the same as ADE except that it gives a strict requirement about how the ADE scheme might increase the distinguishing advantage of the adversary. In particular, M1-ADE requires that the advantage of the adversary in the ADE game can only be an additive factor more than the distinguishing advantage of the original distributions. This requirement allows for “recursively” reinterpreting an application of ADE between two distributions as indistinguishable distributions themselves and reapplying ADE any polynomial number of times. With standard ADE, this could multiplicatively scale the adversary’s advantage, and hence it is only safe to “recurse” a constant number of times. Note that this stronger notion of encryption is also trivially instantiated by RO.

It is not clear how reasonable such an assumption is; however, we found it interesting that the ADS paradigm is even capable of such a thing with a slight modification. We find that the M1-ADE definitions are not as readily composable with other cryptography, on top of being a non-standard definition. As such, we focus the majority of our efforts in this paper exploring standard ADE/ADS. For more information on the technical details regarding M1-ADE, see Appendix B.

1.4 Roadmap of our approach

In the main body of this work we introduce definitions for ADE, ADOT, and ADGC in Section 3 and then we demonstrate a construction pathway to achieve ADOT and 2PC with online-costs $O(n + \text{poly}(\lambda))$ and $2m + O(n + \text{poly}(\lambda))$ respectively. In particular, this pathway mostly consists of simple relational results that generally show the interconnectedness of ADS primitives, which we hope demonstrates the utility and elegance of the ADS framework as a whole.

Pathway. We give the pathway in steps.

- **(Theorem 1)** We first start by showing that there exists a so-called universal ADE scheme that can be built from RO with online-cost $n\lambda$. Technically, this ADE scheme is secure for any opening $\mathcal{X}_n$, but we only require that this result implies the existence of succinct $\{1\}$ -ADE, which only encrypts one message and always ends up opening the message.
- **(Theorem 2)** From $\{1\}$ -ADE and any selectively secure garbling scheme, we construct an ADGC scheme with online-cost $|\tilde{x}| + \text{poly}(\lambda)$ for label size $|\tilde{x}|$ by simply encrypting the garbled circuit $\tilde{C}$ and decoding table d with $\{1\}$ -ADE. We send the ciphertext for that in the offline-phase, and we send the $\{1\}$ -ADE scheme’s succinct key in the online-phase.
- **(Fact 1)** We then combine the above transformation with recent results in the space of arithmetic garbled circuits with good rate [BLLL23] to achieve a garbling scheme with very short keys in the online-phase. Namely, our ADGC scheme assumes decisional composite residuosity (DCR) and has online label size $O(n + \text{poly}(\lambda))$. Note that while the [BLLL23] garbling technique manipulates arithmetic values, our resulting ADGC scheme manipulates Boolean labels.
- **(Theorem 8)** We perform a trivial transformation on the above ADGC scheme to obtain universal ADE for the same online cost: $O(n + \text{poly}(\lambda))$ bits.
- **(Theorem 4)** Then we show that $\mathcal{X}_n^{1/2}$ -ADE can be used to construct $\mathcal{X}_n^{1/2}$ -ADOT, where $\mathcal{X}_n^{1/2}$ is the openings ensemble associated with standard 1-out-of-2 OT, where the receiver chooses between two messages in a batched fashion. This construction makes use of Beaver’s preprocessed OT [Bea95] by running the preprocessing step in the offline-phase, and then encrypting the possible responses for each message using ADE, so that instead of having to send the responses directly in the online-phase, which would be too large, we can instead send a succinct decryption key.
- **(Corollary 1)** We then combine the prior two results to construct an $\mathcal{X}_n^{1/2}$ -ADOT scheme with online-cost $O(n + \text{poly}(\lambda))$, which is succinct in the message sizes.

- **(Theorems 2 and 18)** Again, we give a trivial transformation from ADGC to get adaptive GC schemes by including an output table of necessary size to abide by the [AIKW13]. In particular, we can take any selectively secure GC scheme with projective labels of total size $n\lambda$ and turn it into an adaptive GC scheme with label cost $n\lambda$ and online-cost m .
- **(Corollary 2)** Lastly, we can use the above ADOT scheme and the above adaptive GC scheme to get 2PC with online-cost $2m + O(n + \text{poly}(\lambda))$ using the ADOT scheme in place of an OT scheme for the normal 2PC from garbled circuits construction.

Framework. Each step in the above pathway consists of a quite simple proof, but in combination we can achieve state-of-the-art results from $\{1\}$ -ADE and DCR using only general transformational theorems. Figure 1 depicts the relationships between ADS objects described by our theorems. Note that for space reasons and because many of them are straightforward, many of the depicted theorems can be found in the appendix.

1.5 Prior Work

Preprocessed OT. There is considerable work in finding OT protocols that do not require the use of expensive cryptography in the time-critical online-phase of the protocol, first posited in [Bea95], who pointed out that random OT could be used in the preprocessing step to give a cryptography-free online-phase protocol. This idea was expanded upon by [IKNP03] giving a technique called OT extension, in which a small number of OTs in either preprocessing- or online-phases could be extended to an arbitrary number of real OTs using symmetric-key primitives. There also exist an extensive literature on adaptive k out of n OT schemes [CNS07, GH08, GH11, JL09, KNP10, KNP11, LLM⁺17, RKP09, Zha11], that achieves near $O(k)$ online-phase cost. Every one of the above OT protocols sends something the size of the received messages in the online-phase, so batched 1-out-of-2 OT costs $O(nm)$ bits at least.

Lower-Bounds. It was shown in [AIKW13] that adaptive simulation-based *randomized encodings*—a more general version of garbling scheme Definition 5—must send something in the online-phase at least the size of the output of the circuit for general circuits. Further, [HW15] gave a similar lower-bound for adaptive simulation-based 2PC against malicious adversaries. In particular, OT is an example of a 2PC functionality that suffers from the [HW15] lower-bound, even when the messages are known by the sender in the offline-phase. [KKPW21] also demonstrated that the naive adaptive version of Yao’s garbled circuit technique, in which the Yao-style garbled circuit is sent in the offline-phase, cannot be shown secure from CPA secure encryption without modification.⁹ Likewise, the works of [BHKO23, GYW⁺23] showed that the lower-bound of [AIKW13] does apply to the NPROM, but not to protocols in the programmable ROM (PROM) [FLR⁺10], giving another separation result between the two models.

Adaptive Garbled Circuits. Garbled circuits seem like a prime candidate for preprocessing, since the vast majority of the protocol material—the garbled circuit itself—can be generated independently of the inputs of the parties. In fact, some work prior to 2012 used selectively-secure garbling schemes adaptively in an attempt to get one-time programs [GGP10, GKR08]; however, [BHR12a] pointed out that selectively secure garbling schemes are not adaptive by default. Further, [AIKW13] later gave a lower-bound that simulation-based adaptive garbled circuits must send something at least the size of the output of the circuit in the online-phase. A similar lower-bound is given in [HW15] for 2PC, which states that maliciously-secure simulation-based adaptive 2PC schemes must also send something at least the size of the output of the circuit in the online-phase. In particular, batched 1-out-of-2 OT is an example of a worst-case functionality by [HW15].

⁹More precisely, there is a necessary security loss that is exponential in the depth of the circuit being garbled.

Weaker Adaptive Garbling Notions. Weaker definitions for adaptive garbling have been proposed as well to get around the lower-bound of [AIKW13] while maintaining some notion of security. The first of these was adaptive indistinguishable garbling posed by [JW16], which is secure only for adversaries that first choose two circuits C_0 and C_1 in the offline-phase, and then choose inputs x_0 and x_1 in the online-phase such that $C_0(x_0) = C_1(x_1)$. Standard model constructions of indistinguishability-based security that circumvent the lower-bound of [AIKW13] are only known for log-depth circuits. Recently, [ABK⁺23] posed a framework for a hybrid distributional and simulation based security notion that is plausibly not subject to the [AIKW13] lower-bound; however, no standard model constructions are posited. Unfortunately, neither weakening is known to be able to construct simulation-based adaptive security, even when the necessary one-time pad the size of the output message is given. We point out that these results are important for demonstrating that there is hope to circumvent the lower-bound of [AIKW13].

Adaptive Garbling from RO. The first garbling scheme to be shown adaptively secure is that of [BHR12a] in the programmable ROM and achieves an online-phase of size $O(n\lambda)$ for number of inputs n by using the RO to one-time pad the garbled circuit.¹⁰ Work by [AIKW13] also gives a protocol for achieving online-cost $n + o(n)$ in the PROM from various public-key assumptions (DDH, LWE, or RSA) for randomized encodings, which is near enough to garbling that we mention it here. The $n + o(n)$ scheme was improved in [DLL24] to also allow for reuse based on PROs and ring-LWE. Recently, [BHKO23, GYW⁺23] showed that many popular garbling schemes are adaptively secure from in the NPROM without modification.¹¹ The online-phase for protocols using [BHKO23, GYW⁺23] scale only with the size of the input labels and decoding table, which must abide by the [AIKW13] lower-bound; however, the works of [AIKW13, BHR12a, DLL24] circumvent the lower-bound using the programmable RO.

Adaptive Garbling from the Standard Model. There is a large variety of standard model adaptive garbling schemes. Starting with results from OWFs, the work of [HJO⁺16] achieves an online-phase of $O(w\lambda)$ for maximum circuit width w from a primitive called somewhere equivocal encryption. Further, [JW16] showed that the naive adaptive version of Yao’s garbling scheme *is* adaptively secure for log-depth circuits. From public-key assumptions, [AS16] achieves an online-phase of size $O(n\lambda + m)$ for output-size m from indistinguishability obfuscation. The state of art for near-optimal garbling is given from the work [GOS18, GS18], which achieves an online-phase of size $O(n + m + \log C)$ from laconic OT—which is known from CDH. None of the above schemes are considered to be viable in practice, due either to concrete efficiency concerns [AS16, GS18, GOS18, HJO⁺16] or for not being able to be applied to a sufficiently expressive circuit class [JW16].

Other Proposed Standard Model Security Definitions. The work of [BHK13] gives a framework for specifying standard model security properties of hash functions, which they call the universal computational extractor (UCE) framework. [BHK13] gives four security definitions under the UCE framework, as well as constructions for many primitives under those definitions, including a construction for simulation-based adaptive garbled circuits with online-phase at least the output size. [BHK13] states that their UCE framework falls into the category of *second-degree assumptions*, which are assumptions that are not phrased as an interactive game between a challenger and an adversary, in contrast to most standard cryptographic assumptions, such as OWFs and indistinguishability obfuscation (iO), which are first-degree. In UCE, the adversary is split into two phases a “source” and a “distinguisher”. The source phase runs first, and it may pass “leakage” to the distinguisher, but, crucially, this leakage is constrained by the UCE security definitions. Thus, while this framework is not expressible as a standard game-based definition, it is still falsifiable, in contrast to models such as NPROM for which we simply take a hash function to behave as an NPRO. Follow-on work [ST17] applied the UCE methodology to the context of pseudorandom permutations.

¹⁰The protocol given in [BHR12a] is garbling-scheme agnostic, and so using better protocols may yield better online-phases.

¹¹[GYW⁺23] shows a stronger cryptanalytic claim that using a popular fixed-key AES based hash function does not lose a factor square root security when fixed key AES is modeled as a non-programmable invertible permutation.

While our goal in this work is not just to introduce a single encryption definition, ADE is a main focus. At first glance, UCE and ADE are quite similar. For instance, the authors of [BHK13] explicitly give results in the input-adaptive setting. Further, the formal definition of $\{1\}$ -ADE presented in this work is also a second-degree assumption, in that ADE schemes require that the indistinguishable distributions they act over are *a priori* indistinguishable, which cannot be empirically tested in a game.

However in terms of their impact, UCE and ADE are incomparable: UCE obtains security for a broad class of interesting problems, including for some that are known to be uninstantiable from first-degree assumptions [Wic13]. This rules out the possibility of instantiating UCE from any combination of first-degree assumptions. For instance, even OWFs and iO are not sufficient to achieve UCE. In contrast, ADE is focused on the specific class of input-adaptive problems. This focus allows us to aim for a weaker security definition, which is more plausibly instantiable. In particular, the first-degree-uninstantiable results from UCE do not seem to be achievable from ADE.

2 Preliminaries

This section presents our general notation as well as formal definitions of existing objects, including OT, garbling schemes, and 2PC. The main non-standard concept here is that of an *opening family* (Notation 2); other definitions are relatively standard, and we invite the reader to refer to them as needed.

2.1 Notation and Concepts

Notation 1 (General Notation).

- $X \approx Y$ denotes that distributions X and Y are computationally indistinguishable.
- $\mathcal{O}$ denotes a random oracle.
- $a, b, \tau \leftarrow \langle A(x), B(y) \rangle$ denotes that interactive processes A and B act on respective inputs x and y . As a result of the interaction, A outputs a and B outputs b . The third output τ denotes the protocol transcript.
- $R \leftarrow_{\$} \{0, 1\}^\infty$ denotes a random tape that is allowed to be any length. In particular, all of the randomized procedures we consider should only require at most polynomial amounts of randomness, so we write $\{0, 1\}^\infty$ as a standin for whatever that amount should be.

Because adaptive security depends on opening subsets of encryptions, many definitions will be associated with a family of valid openings $\mathcal{X}_n$ for associated number of messages n . As such, we will use the following notation.

Notation 2 (Opening Family). An opening family $\{\mathcal{X}_n\}_{n \in \mathbb{N}}$ is a set ensemble where each set $\mathcal{X}_n \subset \{0, 1\}^n$ corresponds to a valid way to open a message vector of length n , such that for $x \in \mathcal{X}_n$, $x_i = 1$ corresponds to the i -th message in a message vector being opened and vice versa for $x_i = 0$. The opened message vector can also be written $\mathbf{m}_x = \{\mathbf{m}_i\}_{x_i=1}$ and we also assume that $x \in \mathcal{X}_n$ can be checked efficiently in the length of x .

We assume that every opening set includes the element 0^n , which corresponds to no opening, i.e. an abort. As a slight abuse of notation and when clear from context, we will simply write $\mathcal{X}_n$ in place of $\{\mathcal{X}_n\}_{n \in \mathbb{N}}$.

Notation 3 (1-out-of-2 Opening Ensemble). We give a common opening ensemble that we will use in this work. A vector 1-out-of-2 selection is an opening ensemble in which for each pair of messages, one chooses to open exactly 1 element of the pair. In regex, the ensemble is described as $\{01, 10\}^{n/2}$. We denote a 1-out-of-2 selection by $\mathcal{X}_n^{1/2}$.

2.2 Primitive Definitions

Definition 1 ($\mathcal{X}_n$ -Oblivious Transfer). An **oblivious transfer** scheme (OT) w.r.t. openings $\mathcal{X}_n$ consists of 4 potentially interactive PPT procedures $\Pi = (\text{Send}_0, \text{Recv}_0, \text{Send}_1, \text{Recv}_1)$ that satisfy the following condition.

- **Correctness:** For all parameters $\lambda, m, n \in \mathbb{N}$, messages $\mathbf{m} \in \{0, 1\}^{n \times m}$, openings $x \in \mathcal{X}_n$, and the following

$$\begin{aligned} s_0, r_0, - &\leftarrow \langle \text{Send}_0(1^\lambda, \mathbf{m}), \text{Recv}_0(1^\lambda, 1^m, 1^n) \rangle \\ -, \mathbf{m}', - &\leftarrow \langle \text{Send}_1(s_0), \text{Recv}_1(r_0, x) \rangle \end{aligned}$$

then it should be that $\mathbf{m}' = \mathbf{m}_x$.

- **Online Communication Complexity:** For the following

$$\begin{aligned} s_0, r_0, - &\leftarrow \langle \text{Send}_0(1^\lambda, \mathbf{m}), \text{Recv}_0(1^\lambda, 1^m, 1^n) \rangle \\ -, -, \tau_1 &\leftarrow \langle \text{Send}_1(s_0), \text{Recv}_1(r_0, x) \rangle \end{aligned}$$

let T be such that $|\tau_1| \leq T(\lambda, m, n)$ for all $\mathbf{m} \in \{0, 1\}^{n \times m}$ and $x \in \mathcal{X}_n$. We say that T is the online communication complexity of Π , and Π is succinct when $T(\lambda, m, n) = o(nm)$, but can scale with λ arbitrarily.

Remark 1. We give a very general definition for OT that any OT scheme with preprocessing may be able to conform to. The various phases are not meant to prescribe any round complexity.

Remark 2. The definition for OT above is not the usual definition since the messages the sender has must be known in the offline-phase. This is because it is information-theoretically impossible to have a scheme in which the sender only knows the messages in the online-phase but maintains succinctness.

Definition 2 ($\mathcal{X}_n$ -Adaptive Simulation Oblivious Transfer Security). An $\mathcal{X}_n$ -OT scheme as in Definition 1 is (**adaptively**) **simulation secure** if it satisfies the following conditions.

- **Simulation Receiver Privacy:** There exists a PPT simulator $\mathcal{S}$ such that for all parameters¹² $m, n \in \mathbb{N}$, messages $\mathbf{m} \in \{0, 1\}^{n \times m}$, inputs $x \in \mathcal{X}_n$, and for the following

$$\begin{aligned} R_0, R_1 &\leftarrow_{\$} \{0, 1\}^\infty \\ s_0, r_0, \tau_0 &\leftarrow \langle \text{Send}_0(1^\lambda, \mathbf{m}; R_0), \text{Recv}_0(1^\lambda, 1^m, 1^n) \rangle \\ -, -, \tau_1 &\leftarrow \langle \text{Send}_1(s_0; R_1), \text{Recv}_1(r_0, x) \rangle \end{aligned}$$

then it should be that

$$\{\mathbf{m}, R_0, R_1, \tau_0, \tau_1\} \approx \{\mathcal{S}(1^\lambda, 1^m, 1^n, \mathbf{m})\}$$

- **Adaptive Simulation Sender Privacy:** There exists a PPT simulator $\mathcal{S} = (\mathcal{S}_1, \mathcal{S}_2)$ such that for all parameters $m, n \in \mathbb{N}$, messages $\mathbf{m} \in \{0, 1\}^{n \times m}$, for all PPT adversaries¹³ $\mathcal{A}$, and for the following games

$\text{Real}_{\Pi, \mathcal{A}, \mathbf{m}}^{\text{sim-ot}}(\lambda)$	
1:	$R_0, R_1 \leftarrow_{\$} \{0, 1\}^\infty$
2:	$s_0, r_0, \tau_0 \leftarrow \langle \text{Send}_0(1^\lambda, \mathbf{m}), \text{Recv}_0(1^\lambda, 1^m, 1^n; R_0) \rangle$
3:	$x, S \leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, R_0, \tau_0)$
4:	$-, -, \tau_1 \leftarrow \langle \text{Send}_1(s_0), \text{Recv}_1(r_0, x; R_1) \rangle$
5:	return (S, R_1, τ_1)

¹²The parameters m and n should be a polynomial in λ .

¹³Adversaries should always output some $x \in \mathcal{X}_n$.

$\text{Ideal}_{\Pi, \mathcal{A}, \mathcal{S}, \mathbf{m}}^{\text{sim-ot}}(\lambda)$	
1 :	$x, S \leftarrow \mathcal{S}_1^{\mathcal{A}}(1^\lambda, 1^m, 1^n)$
2 :	if $x \in \mathcal{X}_n$:
3 :	return $\mathcal{S}_2^{\mathcal{A}}(S, \mathbf{m}_x)$
4 :	else :
5 :	return $\perp$

then it should be the case that

$$\{\text{Real}_{\Pi, \mathcal{A}, \mathbf{m}}^{\text{sim-ot}}(\lambda)\} \approx \{\text{Ideal}_{\Pi, \mathcal{A}, \mathcal{S}, \mathbf{m}}^{\text{sim-ot}}(\lambda)\}$$

Definition 3 (Garbled Circuits). A **garbling** scheme is a tuple of PPT algorithms:

$$\Pi = (\text{Garble}, \text{Encode}, \text{Eval}, \text{Decode})$$

Π must satisfy the following correctness condition:

- **Correctness:** For all parameters $\lambda, m, n \in \mathbb{N}$, inputs $x \in \{0, 1\}^m$, and circuits $C : \{0, 1\}^n \rightarrow \{0, 1\}^m$, and the following

$$\begin{aligned} \tilde{C}, e, d &\leftarrow \text{Garble}(1^\lambda, C) \\ \tilde{x} &\leftarrow \text{Encode}(e, x) \\ \tilde{y} &\leftarrow \text{Eval}(\tilde{C}, \tilde{x}) \\ y &\leftarrow \text{Decode}(d, \tilde{y}) \end{aligned}$$

then it should be the case that $y = C(x)$.

- **Online Communication Complexity:** For the following

$$\begin{aligned} \tilde{C}, e, d &\leftarrow \text{Garble}(1^\lambda, C) \\ \tilde{x} &\leftarrow \text{Encode}(e, x) \end{aligned}$$

let T be such that $|d| + |\tilde{x}| \leq T(\lambda, C)$ for adaptive garbling scheme (AGC) (Definition 5) or $|d| + |\tilde{x}| + |\tilde{C}| \leq T(\lambda, C)$ for selective garbling scheme (SGC) (Definition 4) for all circuits $C : \{0, 1\}^n \rightarrow \{0, 1\}^m$. We say that T is the online communication complexity of Π , and that Π is circuit-succinct if $T(\lambda, C) = o(|C|)$ and succinct if $T(\lambda, C) = o(m)$.¹⁴ Π can be succinct and scale with λ arbitrarily.

Definition 4 (Selective Simulation Garbled Circuit Security). A garbled circuit scheme Π as in Definition 3 can be selectively simulation secure in the manner shown below.

- **Selective Simulation Privacy:** There exists a PPT simulator $\mathcal{S}$ such that for all parameters¹⁵ $m, n \in \mathbb{N}$, for all circuits $C : \{0, 1\}^n \rightarrow \{0, 1\}^m$, for all inputs $x \in \{0, 1\}^n$, for all PPT adversaries $\mathcal{A}$, and for the following sampling procedure

$$\begin{aligned} \tilde{C}, e, d &\leftarrow \text{Garble}(1^\lambda, C) \\ \tilde{x} &\leftarrow \text{Encode}(e, x) \end{aligned}$$

it should be that

$$\{\tilde{C}, \tilde{x}, d\} \approx \{\mathcal{S}(1^\lambda, \Phi(C), C(x))\}$$

¹⁴The lower-bound for online communication complexity of simulation-based garbling as in Definition 5 is the size of the output of the circuit, shown in [AIKW13], which makes this particular definition of succinctness interesting to beat.

¹⁵The parameters m and n and the size of circuit C should be a polynomial in λ .

for topology of circuit $\Phi(C)$.

Definition 5 (Adaptive Simulation Garbled Circuit Security). A garbled circuit scheme Π as in Definition 3 can be **adaptively simulation secure** in the manner shown below.

- **Adaptive Simulation Privacy:** There exists a PPT simulator $\mathcal{S} = (\mathcal{S}_1, \mathcal{S}_2)$ such that for all parameters $m, n \in \mathbb{N}$, for all circuits $C : \{0, 1\}^n \rightarrow \{0, 1\}^m$, for all PPT adversaries $\mathcal{A}$, and for the following games

$\text{Real}_{\Pi, \mathcal{A}, C}^{\text{sim-prv}}(\lambda)$	$\text{Ideal}_{\Pi, \mathcal{A}, \mathcal{S}, C}^{\text{sim-prv}}(\lambda)$
1: $\tilde{C}, e, d \leftarrow \text{Garble}(1^\lambda, C)$	1: $\tilde{C}, \tilde{x}, S_S \leftarrow \mathcal{S}_1(1^\lambda, \Phi(C))$
2: $x, S \leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, \tilde{C})$	2: $x, S_A \leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, \tilde{C})$
3: $\tilde{x} \leftarrow \text{Encode}(e, x)$	3: $d \leftarrow \mathcal{S}_2(S_S, C(x))$
4: return $(S, \tilde{x}, d)$	4: return $(S_A, \tilde{x}, d)$

for topology of circuit $\Phi(C)$, then it should be that

$$\{\text{Real}_{\Pi, \mathcal{A}, C}^{\text{sim-prv}}(\lambda)\} \approx \{\text{Ideal}_{\Pi, \mathcal{A}, \mathcal{S}, C}^{\text{sim-prv}}(\lambda)\}$$

Definition 6 (2-Party Computation). A **2 party computation** scheme (2PC) is a tuple of 4 potentially interactive PPT procedures $\Pi = (\text{Offline}_0, \text{Offline}_1, \text{Online}_0, \text{Online}_1)$ with the following properties.

- **Correctness:** For all parameters $\lambda, m, n \in \mathbb{N}$, circuits $C : \{0, 1\}^{n_0} \times \{0, 1\}^{n_1} \rightarrow \{0, 1\}^m$ such that $n_0 + n_1 = n$, all inputs $x_0 \in \{0, 1\}^{n_0}$ and $x_1 \in \{0, 1\}^{n_1}$, and for the following

$$s_0, s_{1,-} \leftarrow \langle \text{Offline}_0(1^\lambda, C), \text{Offline}_1(1^\lambda, C) \rangle$$

$$y'_0, y'_1, - \leftarrow \langle \text{Online}_0(s_0, x_0), \text{Online}_1(s_1, x_1) \rangle$$

then it should be that $y'_0 = y_0$ and $y'_1 = y_1$ for $(y_0, y_1) = C(x_0, x_1)$.

- **Online Communication Complexity:** For the following

$$s_0, s_{1,-} \leftarrow \langle \text{Offline}_0(1^\lambda, C), \text{Offline}_1(1^\lambda, C) \rangle$$

$$-, -, \tau_1 \leftarrow \langle \text{Online}_0(s_0, x_0), \text{Online}_1(s_1, x_1) \rangle$$

let T be such that $|\tau_1| \leq T(\lambda, C)$ for all circuits $C : \{0, 1\}^{n_0} \times \{0, 1\}^{n_1} \rightarrow \{0, 1\}^m$ such that $n_0 + n_1 = n$ and all inputs $x_0 \in \{0, 1\}^{n_0}$ and $x_1 \in \{0, 1\}^{n_1}$. We say that T is the online communication complexity of Π , and that Π is circuit-succinct if $T(\lambda, C) = o(|C|)$ and succinct if $T(\lambda, C) = o(m)$.¹⁶ Π can be succinct and scale with λ arbitrarily.

Definition 7 (Adaptive Simulation 2PC). A 2PC scheme as in Definition 6 is **adaptively simulation secure** if it satisfies the following condition for both parties.

- **Simulation P_b Privacy:** There exists a PPT simulator $\mathcal{S} = (\mathcal{S}_1, \mathcal{S}_2)$ such that for all parameters¹⁷ $m, n \in \mathbb{N}$, circuits $C : \{0, 1\}^{n_0} \times \{0, 1\}^{n_1} \rightarrow \{0, 1\}^m$, inputs $x_b \in \{0, 1\}^{n_b}$, and for the following games

$\text{Real}_{\Pi, \mathcal{A}, C, x_b}^{b\text{-sim-2pc}}(\lambda)$
1: $R_0, R_1 \leftarrow \{0, 1\}^\infty$
2: $s_b, s_{1-b}, \tau_0 \leftarrow \langle \text{Offline}_b(1^\lambda, C), \text{Offline}_{1-b}(1^\lambda, C; R_0) \rangle$
3: $x_{1-b}, S \leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, R_0, \tau_0)$
4: $-, -, \tau_1 \leftarrow \langle \text{Online}_b(s_b, x_b), \text{Online}_{1-b}(s_{1-b}, x_{1-b}; R_1) \rangle$
5: return (S, R_1, τ_1)

¹⁶The lower-bound of [HW15] shows that malicious simulation-based security cannot achieve better than $\Omega(m)$ for some circuits in the online-phase, which makes this particular definition of succinctness an interesting target to beat.

¹⁷The parameters m and n and the size of circuit C should be a polynomial in λ .

$\text{Ideal}_{\Pi, \mathcal{A}, \mathcal{S}, C, x_b}^{b\text{-sim-2pc}}(\lambda)$	
1 :	$x_{1-b}, S_S \leftarrow S_1^{\mathcal{A}}(1^\lambda)$
2 :	return $S_2^{\mathcal{A}}(S_S, C(x_0, x_1))$

then it should be the case that

$$\{\text{Real}_{\mathcal{A}, C, x_b}^{b\text{-sim-2pc}}(\lambda)\} \approx \{\text{Ideal}_{\mathcal{A}, \mathcal{S}, C, x_b}^{b\text{-sim-2pc}}(\lambda)\}$$

3 A New Framework: Adaptive Distributional Security

This section formalizes the main ideas of the ADS framework and applies ADS to various primitives.

We first introduce a central definition: $\mathcal{X}_n$ -indistinguishability. Normally, two distributions are indistinguishable if the full outputs of the distributions are indistinguishable. In the following definition, distributions output side information σ , efficient side information function ξ , and messages $\mathbf{m}$ that can only be opened according to $x \in \mathcal{X}_n$. The distributions should be indistinguishable when some adversary is first given σ , then produces an opening $x \in \mathcal{X}_n$ and side choice η , and finally sees $\mathbf{m}_x$ and $\xi(x, \eta)$.

Ultimately, our goal is to introduce a *composable* framework for security definitions related to input-adaptivity, and the side information σ and ξ act as our framework's glue. In particular, σ and ξ give a formal context by which a security definition can capture the abstract notion of the adversary's view of the past—namely σ —and the adversary's view of the future—namely ξ . As an example of this, consider an adaptive garbling scheme, which sends a garbled circuit in advance (hence, $\sigma = \tilde{C}$), then the evaluator decides their input and gets corresponding labels (hence, $\mathbf{m}_x = \tilde{x}$), and finally the evaluator receives the decoding table ($\xi(x, \eta) = d$). Using the notion of $\mathcal{X}_n$ -indistinguishability, we can talk abstractly about many different adaptive cryptographic systems in a distributional manner:

Definition 8 ($\mathcal{X}_n$ -Indistinguishable Distributions). *We say that—not necessarily PPT—distributions D_0 and D_1 are **indistinguishable w.r.t. openings** $\mathcal{X}_n$ if for all¹⁸ $n, m \in \mathbb{N}$ and for all PPT adversaries $\mathcal{A}$*

$\text{Game}_{\mathcal{A}, D}^{\text{d-ind}}(\lambda, m, n)$	
1 :	$\sigma, \xi, \mathbf{m} \leftarrow_{\$} D(\lambda, m, n)$
2 :	$x, \eta, S \leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, \sigma)$
3 :	if $x \notin \mathcal{X}_n$:
4 :	$x \leftarrow 0^n$
5 :	return $(S, \xi(x, \eta), \mathbf{m}_x)$

$$\{\text{Game}_{\mathcal{A}, D_0}^{\text{d-ind}}(\lambda, m, n)\} \approx \{\text{Game}_{\mathcal{A}, D_1}^{\text{d-ind}}(\lambda, m, n)\}$$

for PPT program ξ .¹⁹

Remark 3. *Technically, the above definition does not explicitly need $\mathbf{m}$ to be present in the view since this information can be baked into ξ to output $\mathbf{m}_x$ along with other side information. We make explicit $\mathbf{m}$ for simplicity of notation.*

We also give a notion of distributional indistinguishability for distributions that output a circuit that is evaluated, instead of distributions that only open messages.

Definition 9 (Indistinguishable Circuit Distributions). *We say that distributions D_0 and D_1 are **indistinguishable circuit distributions** if for all PPT adversaries $\mathcal{A}$*

¹⁸The parameters m and n should be a polynomial in λ .

¹⁹Note that the size of σ and ξ should be globally bounded by a polynomial in λ, m, n per distribution.

$\text{Game}_{\mathcal{A},D}^{c\text{-ind}}(\lambda, m, n)$
1: $\sigma, \xi, C \leftarrow \$ D(\lambda, m, n)$
2: $x, \eta, S \leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, \sigma)$
3: return $(S, \xi(x, \eta), C(x))$

$$\{\text{Game}_{\mathcal{A},D_0}^{c\text{-ind}}(\lambda, m, n)\} \approx \{\text{Game}_{\mathcal{A},D_1}^{c\text{-ind}}(\lambda, m, n)\}$$

for poly-sized circuits $C : \{0, 1\}^n \rightarrow \{0, 1\}^m$ with the same topology²⁰ when output by either D_0 or D_1 , and PPT program ξ .²¹

Now, we formalize a central primitive: adaptive distributional encryption (ADE). ADE is an encryption scheme that prevents an adversary from breaking some other ongoing processes that might be related to whatever is being encrypted, even when we give a decryption key to the adversary later. In contrast, most off-the-shelf standard model encryption schemes tend to lose their security guarantees if the secret key is ever revealed. We capture this security notion by giving an interactive game over $\{1\}$ -indistinguishable distributions— $\mathcal{X}_n$ -indistinguishable distributions in which the single message is always opened—and requiring that any such distributions maintain indistinguishability even when the adversary is given an encryption of the single message followed in the online-phase by a decryption key. Note that we are particularly interested in distributions that *are easily* distinguishable when given the messages right away, but still satisfy $\mathcal{X}_n$ -indistinguishability.

Definition 10 (Adaptive Distributional Encryption). An **adaptive distributional encryption** scheme (ADE) is an encryption scheme $\Pi = (\text{Enc}, \text{Dec})$ with the following properties.

- **Correctness:** For all parameters $\lambda, m \in \mathbb{N}$, messages $\bar{m} \in \{0, 1\}^m$, and for

$$\begin{aligned} c, k &\leftarrow \text{Enc}(1^\lambda, \bar{m}) \\ \bar{m}' &\leftarrow \text{Dec}(k, c) \end{aligned}$$

it should be that $\bar{m}' = \bar{m}$.

- **Security:** For all parameters $m \in \mathbb{N}$, $\{1\}$ -indistinguishable distributions D_0 and D_1 , for all PPT adversaries $\mathcal{A}$, and for game

$\text{Game}_{\Pi, \mathcal{A}, D}^{\text{ade}}(\lambda, m)$
1: $\sigma, \xi, \bar{m} \leftarrow \$ D(\lambda, m, 1)$
2: $c, k \leftarrow \text{Enc}(1^\lambda, \bar{m})$
3: $\eta, S \leftarrow \mathcal{A}(1^\lambda, \sigma, c)$
4: return $(S, \xi(\eta), k)$

it should be that

$$\{\text{Game}_{\Pi, \mathcal{A}, D_0}^{\text{ade}}(\lambda, m)\} \approx \{\text{Game}_{\Pi, \mathcal{A}, D_1}^{\text{ade}}(\lambda, m)\}$$

- **Online Succinct:** For the following

$$-, k \leftarrow \text{Enc}(1^\lambda, \bar{m})$$

let T be such that $|k| \leq T(\lambda, m)$ for all $\bar{m} \in \{0, 1\}^m$. We say that T is the online complexity of Π , and Π is succinct when $T(\lambda, m) = o(m)$, but can scale with λ arbitrarily.

²⁰The specific definition of topology can depend on the protocol using the notion.

²¹Note that the sizes of σ , ξ , and C should be globally bounded by a polynomial in λ, m, n per distribution.

We also give an extension to ADE that applies to encrypting a set of messages and then opening some subset.

Definition 11 ($\mathcal{X}_n$ -Adaptive Distributional Encryption). *An adaptive distributional encryption scheme w.r.t. openings $\mathcal{X}_n$ consists of 3 PPT protocols $\Pi = (Enc, Open, Dec)$ with the following properties.*

- **Correctness:** For all parameters $\lambda, m, n \in \mathbb{N}$, messages $\mathbf{m} \in \{0, 1\}^{n \times m}$, openings $x \in \mathcal{X}_n$, and the following

$$\begin{aligned} c, k &\leftarrow Enc(1^\lambda, \mathbf{m}) \\ k_x &\leftarrow Open(k, x) \\ \mathbf{m}' &\leftarrow Dec(k_x, x, c) \end{aligned}$$

then it should be the case that $\mathbf{m}' = \mathbf{m}_x$. If $x \notin \mathcal{X}_n$, then *Open* should treat x as if it is 0^n .

- **Security:** For all parameters $m, n \in \mathbb{N}$, $\mathcal{X}_n$ -indistinguishable distributions D_0 and D_1 , for all PPT adversaries $\mathcal{A}$, and for game²²

Game _{$\Pi, \mathcal{A}, D$} ^{ade} (λ, m, n)	
1:	$\sigma, \xi, \mathbf{m} \leftarrow \$ D(\lambda, m, n)$
2:	$c, k \leftarrow Enc(1^\lambda, \mathbf{m})$
3:	$x, \eta, S \leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, \sigma, c)$
4:	$k_x \leftarrow Open(k, x)$
5:	return $(S, \xi(x, \eta), k_x)$

it should be that

$$\{\text{Game}_{\Pi, \mathcal{A}, D_0}^{\text{ade}}(\lambda, m, n)\} \approx \{\text{Game}_{\Pi, \mathcal{A}, D_1}^{\text{ade}}(\lambda, m, n)\}$$

- **Online Succinct:** For the following

$$\begin{aligned} _, k &\leftarrow Enc(1^\lambda, \mathbf{m}) \\ k_x &\leftarrow Open(k, x) \end{aligned}$$

let T be such that $|k_x| \leq T(\lambda, m, n)$ for all $\mathbf{m} \in \{0, 1\}^{n \times m}$ and $x \in \mathcal{X}_n$. We say that T is the online complexity of Π , and Π is succinct when $T(\lambda, m, n) = o(nm)$, but can scale with λ arbitrarily.

Remark 4. If a protocol is an $\mathcal{X}_n$ -ADE for any $\mathcal{X}_n$, we will say that it is a universal ADE scheme.

Remark 5. From now on we will refer to the ADE scheme of Definition 10 as $\{1\}$ -ADE, since it can be viewed as an $\mathcal{X}_n$ -ADE scheme where there is only one message and it is opened.

The OT and GC variant security definitions are similar in structure to their simulation-based counterparts, but with intuitive ADS-style definitions that require indistinguishability over $\mathcal{X}_n$ - or circuit-indistinguishable distributions instead of over all message or all circuits, as it is in simulation-based security.

Definition 12 ($\mathcal{X}_n$ -Adaptive Distributional Oblivious Transfer Security). *An $\mathcal{X}_n$ -OT scheme Π as in Definition 1 is **adaptively distributionally secure** if it satisfies the following conditions.*

- **Adaptive Simulation Receiver Privacy:** Same as (adaptive) simulation receiver privacy as in Definition 2.

²²The following game has the same name as the one used in Definition 10 since this is an extension, we will view this one as the canonical version.

- **Adaptive Distributional Sender Privacy:** For all parameters $m, n \in \mathbb{N}$, $\mathcal{X}_n$ -indistinguishable distributions D_0 and D_1 , for all PPT adversaries²³ $\mathcal{A}$, and for game

$\text{Game}_{\Pi, \mathcal{A}, D}^{\text{adot}}(\lambda, m, n)$
1: $\sigma, \xi, \mathbf{m} \leftarrow_{\$} D(\lambda, m, n)$
2: $R_0, R_1 \leftarrow_{\$} \{0, 1\}^\infty$
3: $s_0, r_0, \tau_0 \leftarrow \langle \text{Send}_0(1^\lambda, \mathbf{m}), \text{Recv}_0(1^\lambda, 1^m, 1^n; R_0) \rangle$
4: $x, \eta, S \leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, \sigma, R_0, \tau_0)$
5: $\rightarrow, \tau_1 \leftarrow \langle \text{Send}_1(s_0), \text{Recv}_1(r_0, x; R_1) \rangle$
6: return $(S, \xi(x, \eta), R_1, \tau_1)$

it should be that

$$\{\text{Game}_{\Pi, \mathcal{A}, D_0}^{\text{adot}}(\lambda, m, n)\} \approx \{\text{Game}_{\Pi, \mathcal{A}, D_1}^{\text{adot}}(\lambda, m, n)\}$$

Definition 13 (Adaptive Distributional Garbled Circuit Security). An **adaptive distributional garbled circuits** scheme (ADGC) consists of 4 PPT protocols $\Pi = (\text{Garble}, \text{Encode}, \text{Eval}, \text{Decode})$ that satisfy the following conditions.

- **Correctness:** Same as correctness in Definition 3.
- **Adaptive Distributional Privacy:** For all indistinguishable circuit distributions D_0 and D_1 , for all PPT adversaries $\mathcal{A}$, and for game

$\text{Game}_{\Pi, \mathcal{A}, D}^{\text{adgc}}(\lambda, m, n)$
1: $\sigma, \xi, C \leftarrow_{\$} D(\lambda, m, n)$
2: $\tilde{C}, e, d \leftarrow \text{Garble}(1^\lambda, C)$
3: $x, \eta, S \leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, \sigma, \tilde{C})$
4: $\tilde{x} \leftarrow \text{Encode}(e, x)$
5: return $(S, \xi(x, \eta), \tilde{x}, d)$

it should be that

$$\{\text{Game}_{\Pi, \mathcal{A}, D_0}^{\text{adgc}}(\lambda, m, n)\} \approx \{\text{Game}_{\Pi, \mathcal{A}, D_1}^{\text{adgc}}(\lambda, m, n)\}$$

4 Constructions

We give the proofs from the pathway that we discussed in Section 1.4.

4.1 Constructing Universal ADE

We first show that the naive construction of a semantically secure encryption scheme from RO is a universal ADE scheme. We give the following construction as a per-message encryption scheme, but a candidate ADE scheme would run each in parallel to encrypt.

²³Adversaries should always output some $x \in \mathcal{X}_n$.

Construction 1 (Universal ADE from ROM). *The implementation from an RO is as follows.*

$$\begin{aligned} \text{Gen}(1^\lambda) &:= \text{sample uniformly from } \{0, 1\}^\lambda \\ \text{Enc}(\mathbf{k}, \overline{m}) &:= \mathcal{O}(\mathbf{k}) \oplus \overline{m} \\ \text{Dec}(\mathbf{k}, c) &:= \mathcal{O}(\mathbf{k}) \oplus c \end{aligned}$$

We then show security by considering that the one-time padded messages are generally information-theoretically hidden unless some key corresponding to an unopened message is found, which can only happen negligibly often.

Theorem 1 (Universal ADE from ROM). *The simple ROM encryption scheme (Construction 1) is also an $\mathcal{X}_n$ -ADE scheme for any opening ensemble $\mathcal{X}_n$ (Definition 11) with online-cost $T(\lambda, m, n) = n\lambda$*

Proof. We first note that Construction 1 is clearly correct for the opening function of simply sending the keys associated with openings, and has size λ keys per encryption, and so the online-cost is $T(\lambda, m, n) = n\lambda$.

Security. To show security we introduce the two following games. Note that we will assume that the distinguisher is given the oracle $\mathcal{O}$ at the end of the protocol. We also give the distribution D and output function ξ oracle access so that the scheme is oracle-composable and so that the scheme is closer to a standard model game, since D and ξ would always be able to depend on any standard model implementation.

$\text{Game}_{\Pi, \mathcal{A}, D}^{\text{ade}}(\lambda, m, n)$	$\text{Hyb}_{\Pi, \mathcal{A}, D}(\lambda, m, n)$
1 : $\sigma, \xi, \mathbf{m} \leftarrow \$ D^\mathcal{O}(\lambda, m, n)$	1 : $\{\mathbf{k}_i \leftarrow \$ \{0, 1\}^\lambda\}_{i \in [n]}$
2 : $\{\mathbf{k}_i \leftarrow \$ \{0, 1\}^\lambda\}_{i \in [n]}$	2 : $\{c_i \leftarrow \$ \{0, 1\}^m\}_{i \in [n]}$
3 : $\{c_i \leftarrow \mathcal{O}(\mathbf{k}_i) \oplus \mathbf{m}_i\}_{i \in [n]}$	3 : $\sigma, \xi, \mathbf{m} \leftarrow \$ D^\mathcal{O}(\lambda, m, n)$
4 : $x, \eta, S \leftarrow \mathcal{A}^\mathcal{O}(1^\lambda, 1^m, 1^n, \sigma, \{c_i\}_{i \in [n]})$	4 : $x, \eta, S \leftarrow \mathcal{A}^\mathcal{O}(1^\lambda, 1^m, 1^n, \sigma, \{c_i\}_{i \in [n]})$
5 : return $(S, \xi^\mathcal{O}(x, \eta), \{\mathbf{k}_i\}_{x_i=1})$	5 : $z \leftarrow \xi^\mathcal{O}(x, \eta)$
	6 : program $\{\mathcal{O}(\mathbf{k}_i) := c_i \oplus \mathbf{m}_i\}_{x_i=1}$
	7 : return $(S, z, \{\mathbf{k}_i\}_{x_i=1})$

Then we are tasked with showing for any $\mathcal{X}_n$ -indistinguishable distributions D_0 and D_1 that

$$\{\text{Game}_{\Pi, \mathcal{A}, D_0}^{\text{ade}}(\lambda, m, n)\} \approx \{\text{Game}_{\Pi, \mathcal{A}, D_1}^{\text{ade}}(\lambda, m, n)\}$$

which we will do in two steps.

First we show that for $b \in \{0, 1\}$,

$$\{\text{Game}_{\Pi, \mathcal{A}, D_b}^{\text{ade}}(\lambda, m, n)\} \approx \{\text{Hyb}_{\Pi, \mathcal{A}, D_b}(\lambda, m, n)\}$$

For the purposes of reaching a contradiction, say that the above games are distinguishable.

First consider only computationally bounded distributions D (that run with time some polynomial in λ, m, n).

We then consider two avenues of attack and show that neither is feasible.

- If we consider what is available to the adversary by line 4, we note that the distributions are only different due to the ciphertexts being drawn differently. In Game the ciphertexts are drawn from the image of the RO while in Hyb the ciphertexts are drawn from uniform. Given only polynomially many queries to the RO, it is not possible to distinguish between the two distributions.
- Excepting the ciphertexts, the adversary can witness the RO being programmed at some point $\mathbf{k}_i$. Of course, this would require that any distinguishing D and $\mathcal{A}$ pair would need to query at least one $\mathbf{k}_i$ before line 6 — in other words, D , $\mathcal{A}$, or ξ would need to query a point $\mathbf{k}_i$ before line 6. Because the keys are drawn uniformly, the probability of this occurring is negligible. Also note that this issue is separate from the prior point since the RO inputs and output images are independent.

Note, however, that the distribution D need not be efficient. Because D is not given the keys k_i or ciphertexts c_i , we can model D as part of the adversary that gets to potentially witness the entirety of the oracle before the game starts, producing a polynomially bounded amount of advice about the RO. This setting is known as the auxiliary-input ROM of [Unr07], and it was shown in [CDGS18, Unr07] to be equivalent within polynomial security loss to a setting in which the adversary gets to program polynomially many locations of the RO prior to the game starting. Because the oracle queries that the challenger makes are uniform, it is only with negligible probability that they will intersect with the adversarially programmed RO locations.

Then the second step is to show

$$\{\text{Hyb}_{\Pi, \mathcal{A}, D_0}(\lambda, m, n)\} \approx \{\text{Hyb}_{\Pi, \mathcal{A}, D_1}(\lambda, m, n)\}$$

In the Hyb games, the adversary does not get any access to $\mathbf{m}$ before they make their decision x, η , and then the distinguisher only sees $\mathbf{m}_x$ and $\xi^{\mathcal{O}}(x, \eta)$ after, where the unopened part of the messages are simply not encoded anywhere. The information that the adversary has access to information theoretically mimics that of the $\mathcal{X}_n$ -indistinguishability games, otherwise the games are identical. As such, the indistinguishability of the Hyb games reduces to that of the $\mathcal{X}_n$ -indistinguishability games for D_0 and D_1 . $\square$

The above proof technique was featured in [BHKO23]. Ample discussion about the technique can be found there.

We then give a conjecture that Construction 1 can be instantiated with AES, followed by discussion.

Construction 2 (Generator from AES). *We give a standard stream pseudorandomness generator from AES as a keyed function $F_k(\cdot)$:*

$$G(s) := F_s(0) || F_s(1) || F_s(2) || \dots$$

for as much pseudorandomness as needed.

Conjecture 1 (AES Generator is a Universal ADE scheme). *Replacing the RO calls $\mathcal{O}(k)$ in Construction 1 with the generator $G(k)$ in Construction 2 is a Universal ADE (Definition 11).*

This conjecture stems from two points. First, the use of ROM in Construction 1 is mild: it uses input keys that are uniform and unstructured. Second, to our knowledge, there are no known attacks against using AES to encrypt n independent messages, then allowing the adversary to open encryption seeds corresponding to some number of the messages. The second point is well studied by works related to selective-opening-attack (SOA) encryption [BHY09, DNRS99, FHKW10, LDL⁺14, ORV14]. If Conjecture 1 is constructively shown to be false for AES, we feel that such an attack would illuminate why security in the adaptive setting is concretely difficult, rather than difficult in the abstract. In light of the above, we are confident that Conjecture 1 makes a reasonable assumption for use in future cryptographic works and as a standard-model-like assumption more generally.

4.2 ADGC from $\{1\}$ -ADE

We show how using $\{1\}$ -ADE to encrypt the garbled material of a selectively secure garbled circuit scheme achieves ADGC security with key-sized online-cost.

Construction 3 (ADGC from $\{1\}$ -ADE and Selectively Secure Garbled Circuits). *The protocol simply encrypts the garbled circuit $\tilde{C}$ and decoding table d together, sending the ciphertext in the offline-phase. Then the garbler sends the decryption key in the online-phase as the new decryption table.*

Note that Construction 3 results in the same scheme proposed in [BHR12a] when $\{1\}$ -ADE is implemented with a programmable RO, however, Construction 3 works for any $\{1\}$ -ADE scheme, not necessarily one in the ROM.

We demonstrate that Construction 3 is an ADGC scheme. We do this by viewing the selectively secure garbled circuit scheme as a pair of $\{1\}$ -indistinguishable distributions, in which D_b is the real execution with the garbled material as the message and label encoding as the output of $\xi(x, \eta)$ for distributionally drawn circuit C_b . We show that D_0 and D_1 are $\{1\}$ -indistinguishable using simulation security, since simulation security is stronger than ADS (see Theorem 18). Then we see that the above is in a form to immediately apply $\{1\}$ -ADE.

Theorem 2 (ADGC from $\{1\}$ -ADE and Garbled Circuits). *Construction 3 is an ADGC scheme (Definition 13) from $\{1\}$ -ADE and a selectively secure garbling scheme (Definitions 4 and 10) with online-cost $T(\lambda, C) = T_{ADE}(\lambda, |\tilde{C}| + |d|, 1)$ for garbled circuit $\tilde{C}$ and decoding table d .*

Proof.

Correctness. The construction simply applies encryption to $\tilde{C}$ and d , which is then revealed on schedule, and so the evaluator has the proper material for evaluating the circuit.

Adaptive Distributional Privacy. Let D_0 and D_1 be indistinguishable circuit distributions. Then consider the following statement using Definition 4 for $\sigma_b, \xi_b, C_b \leftarrow \$ D_b$ for $b \in \{0, 1\}$ and efficient adversary $\mathcal{A}$. For sampling procedure

$$\begin{aligned} x_b, y_b, S_b &\leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, \sigma_b) \\ \tilde{C}_b, e, d_b &\leftarrow \text{Garble}(1^\lambda, C_b) \\ \tilde{x}_b &\leftarrow \text{Encode}(e, x_b) \end{aligned}$$

we have that

$$\begin{aligned} \{\sigma_0, \xi_0(x_0, \eta_0), S_0, \tilde{C}_0, \tilde{x}_0, d_0\} &\approx \{\sigma_0, \xi_0(x_0, \eta_0), S_0, \mathcal{S}(1^\lambda, \Phi(C_0), C_0(x_0))\} \\ &\approx \{\sigma_1, \xi_1(x_1, \eta_1), S_1, \mathcal{S}(1^\lambda, \Phi(C_1), C_1(x_1))\} \approx \{\sigma_1, \xi_1(x_1, \eta_1), S_1, \tilde{C}_1, \tilde{x}_1, d_1\} \end{aligned}$$

The jump from Real to Ideal holds due to selective simulation privacy (Definition 5), since the adversary can internally mind σ_b and $\xi_b(x_b, \eta_b)$, and then jump between the Ideal games is due to the circuit indistinguishability of the distributions D_0 and D_1 .

We can then consider viewing the above garbling scheme as a distribution on which $\{1\}$ -ADE can apply. Consider distributions D'_0 and D'_1 implemented in the following way.

$D'_b(\lambda, m' = \tilde{C} + d , n' = 1)$	
1 :	$\sigma, \xi, C \leftarrow \$ D_b(\lambda, m, n)$
2 :	$\tilde{C}, e, d \leftarrow \text{Garble}(1^\lambda, C)$
3 :	$\xi'(1, (x, \eta)) := (\xi(1, (x, \eta)), \text{Encode}(e, x))$
4 :	$\bar{m} \leftarrow (\tilde{C}, d)$
5 :	return $(\sigma, \xi', \bar{m})$

Then we see that by the real game indistinguishability we showed earlier that the distributions D'_0 and D'_1 are $\mathcal{X}'_n$ -indistinguishable at least for $\mathcal{X}'_1 = \{1\}$. Then we see that we can immediately apply $\{1\}$ -ADE to get an adaptively secure ADGC scheme by encrypting the only element $(\tilde{C}, d)$. $\square$

4.3 Succinct Universal ADE from $\{1\}$ -ADE and DCR

We make use of recent advances in low-rate garbled circuits to build an efficient ADE scheme.

Fact 1 (Constant-Rate Label Decomposition Gadget [BLLL23]). *There exists an **arithmetic garbling scheme** (Definition 5) from DCR that takes in an arithmetic label $\tilde{x}$ of size $O(n + \text{poly}(\lambda))$ corresponding to an input x of length m that computes the functionality $\mathbf{m}_x$ for messages of arbitrary length.*

We use a trivial appendix theorem, Theorem 8, to turn any ADGC scheme into a universal ADE scheme of the same cost in conjunction with Fact 1 and the prior Theorem 2 to get a succinct universal ADE scheme by composing the aforementioned statements.

Theorem 3 (Succinct Universal ADE from $\{1\}$ -ADE and DCR). *There exists Universal ADE from $\{1\}$ -ADE and DCR (Definitions 10 and 11) with online-cost $T(\lambda, m, n) = O(n + \text{poly}(\lambda))$ if $T_{ADE}(\lambda, m, 1) = \text{poly}(\lambda)$.*

Proof. Firstly, Fact 1 gives us a garbling scheme with label cost $|\tilde{x}| = O(n + \text{poly}(\lambda))$. Secondly, we saw from Theorem 2 that given a garbling scheme and $\{1\}$ -ADE, we can encrypt the garbled circuit and decoding table as in [BHR12a] to construct an ADGC scheme with online-phase of size $|\tilde{x}| + \text{poly}(\lambda)$, which becomes $O(n + \text{poly}(\lambda))$ when using the construction from Fact 1. Then we make use of a Theorem 8—which shows that ADGC implies universal ADE by simply ignoring hiding for the evaluator’s choice—to get a universal ADE scheme of the same online-cost $O(n + \text{poly}(\lambda))$. $\square$

4.4 Succinct $\mathcal{X}_n^{1/2}$ -ADOT from $\mathcal{X}_n^{1/2}$ -ADE and DCR

We demonstrate how to get $\mathcal{X}_n^{1/2}$ -ADOT by applying $\mathcal{X}_n^{1/2}$ -ADE as a wrapper over Beaver’s preprocessing OT protocol [Bea95] to get a succinctness guarantee that scales only with the online-cost of the ADE scheme.

Construction 4 ($\mathcal{X}_n^{1/2}$ -ADOT from $\mathcal{X}_n^{1/2}$ -ADE and $\mathcal{X}_n^{1/2}$ -OT). *The protocol first invokes Beaver’s preprocessing OT [Bea95] and the sender encrypts the output options prescribed from [Bea95] using ADE and sends the ciphertexts. Then in the online-phase the sender finishes the scheme from [Bea95] but decrypts the ciphertexts using ADE instead of sending the openings directly. Figure 2 demonstrates the protocol flow.*

Theorem 4 ($\mathcal{X}_n^{1/2}$ -ADOT from $\mathcal{X}_n^{1/2}$ -ADE and $\mathcal{X}_n^{1/2}$ -OT). *Construction 4 is an $\mathcal{X}_n^{1/2}$ -ADOT (Definition 12) from $\mathcal{X}_n^{1/2}$ -ADE and selectively secure $\mathcal{X}_n^{1/2}$ -OT (Definitions 2 and 11)*

Proof. We first note that correctness of the over protocol comes primarily from beaver’s preprocessing OT correctness, since the ADE scheme merely moves the online-phase step to the offline-phase and opens the OT the same way. In particular, $x' \in \mathcal{X}_n^{1/2}$ iff $x \in \mathcal{X}_n^{1/2}$.

Security. If we view Beaver’s preprocessing OT as a distribution where the preprocessing step is encoded in σ and the online phase consists of some messages that appear from the other party (the $\mathbf{m}$ distribution), then we find that applying ADE simply moves the online-phase messages to the offline-phase by definition, since the messages are known in-advance. $\square$

Then by applying Theorem 3 we can use Theorem 4 to get a succinct $\mathcal{X}_n^{1/2}$ from our running succinct universal ADE scheme.

Corollary 1 (Succinct $\mathcal{X}_n^{1/2}$ -ADOT from $\{1\}$ -ADE and DCR). *There exists an $\mathcal{X}_n^{1/2}$ -ADOT (Definition 12) from $\{1\}$ -ADE (Definition 10) and DCR with a constant number of online rounds and online-cost $T(\lambda, m, n) = O(n + \text{poly}(\lambda))$ if $T_{ADE}(\lambda, m, 1) = \text{poly}(\lambda)$.*

Then note that using Theorems 2 and 18, the latter of which gives a trivial construction of AGC from ADGC with the same online-cost with an added m for a one-time-pad decoding table, to get an AGC scheme from any SGC scheme with non-label online-cost m . Then, by applying Theorem 16 we also get the following.

Corollary 2 (Succinct 2PC from $\{1\}$ -ADE and DCR). *There exists 2PC (Definition 7) from $\{1\}$ -ADE and DCR (Definition 10) with a constant number of online rounds and online-cost $T(\lambda, C) = 2m + O(n + \text{poly}(\lambda))$ if $T_{ADE}(\lambda, m, 1) = \text{poly}(\lambda)$.*

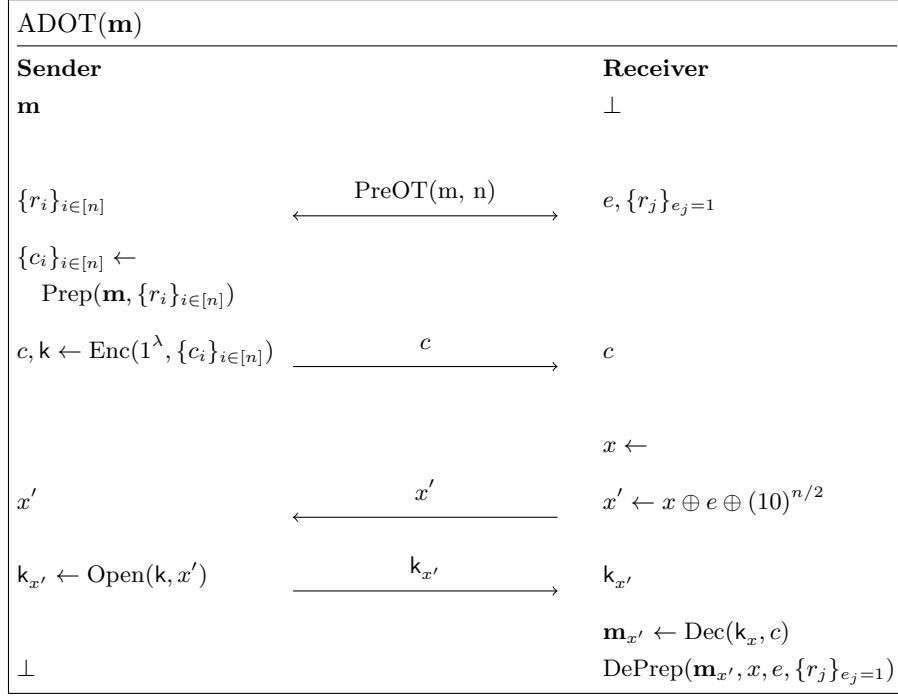


Figure 2: Protocol flow of Construction 4. First recall that n is twice the number of choices, since it is the number of total messages being chosen from. The PreOT step leaves the sender with n one-time pads and the receiver with one from each pair selected by some uniform string e' . e is the expanded version of e' so that $e \in \mathcal{X}_n^{1/2}$ based on choices from e' like a traditional OT scheme. The Prep procedure puts together the messages to send as in the online-phase, preparing for either way that the receiver could choose, by computing the following:²⁴

$$\{c_{i||b} \leftarrow (r_{i||b} \oplus [\mathbf{m}]_{i||0}, r_{i||(1-b)} \oplus [\mathbf{m}]_{i||1})\}_{i \in [n/2], b \in \{0,1\}}$$

And of the output message $\mathbf{m}_{x'}$, only the half of each message decided by x is the decrypted part since the other half is garbage. Hence DePrep computes:

$$\{\overline{m}_j \leftarrow ([\mathbf{m}_{x'}]_j)_a \oplus r_{j||b}\}_{x_{j||a}=1=e_{j||b}}$$

5 Note on Broader Theorems

The statements of Theorem 3 and Corollaries 1 and 2 can be made more broad than is presented in this work. We gave the theorem and corollary statements for $\{1\}$ -ADE schemes with online-cost $T(\lambda, m, 1) = \text{poly}(\lambda)$ to make it more clear where online-cost comes from and to make the chain of theorems easier to follow without needing to get into the specific details of constructions, such as that from [BLLL23] mentioned in Fact 1. In this section we will give the upper-bound for the cost of $\{1\}$ -ADE for which the resulting primitives from Theorem 3 and Corollaries 1 and 2 are still succinct.

- **Theorem 3:** The construction given in Fact 1 mostly scales in cost with the so-called decomposition gadget, which takes as input a compressed arithmetic DCR label and decomposes it into Yao-style labels for each bit in the arithmetic label. The decomposition gadget from [BLLL23] scales in $O(w\kappa + \kappa^3)$ a w -bit label and DCR security parameter κ . As such, as long as $T_{ADE}(\lambda, O(mn\kappa + \kappa^3), 1) = o(mn)$ or rather $T_{ADE}(\lambda, m', 1) = o(m')$ the resulting universal ADE scheme is succinct. The resulting scheme would have cost $T(\lambda, m, n) = O(n + T_{ADE}(\lambda, O(mn\kappa + \kappa^3), 1))$.
- **Corollaries 1 and 2:** The cost of the constructions used to prove these corollaries scales with Fact 1 in the same way. As such we find that an $\{1\}$ -ADE scheme with online-cost $T_{ADE}(\lambda, m, 1) = o(m)$ should suffice. The resulting schemes would have costs

$$T_{ADOT}(\lambda, m, n) = O(n + T_{ADE}(\lambda, O(mn\kappa + \kappa^3), 1))$$

and

$$T_{2PC}(\lambda, m, n) = 2m + O(n + T_{ADE}(\lambda, O(mn\kappa + \kappa^3), 1))$$

respectively.

6 Discussion and Future Work

We presented many constructions in this work that illustrate how interconnected adaptive distributional security is as a framework; however, we feel that there could be even more ways in which the protocols in this work can be more efficient or further reaching. We posit some future directions below.

More ADS Settings. The scope of protocols that we applied the ADS framework to is considerably narrow. We wanted to give meaningful examples of the framework to popular secure computation settings; however, we feel that the general principals explored in this work may be applicable to many areas, in the same way that simulation-based security is ubiquitous.

MPC. We found 2PC to be readily accepting of the ADOT scheme in the place of simulation-based OT (Theorem 16); however, it is not clear how to use the preprocessing step to get a similar scheme for any number of parties. There is already work on the topic of achieving MPC from garbling and OT schemes [BMR90], so it seems like an interesting addition to also consider the case of preprocessing.

Succinct ADE from DDH. We chose to give a succinct protocol for $\mathcal{X}_n^{1/2}$ -ADE from $\{1\}$ -ADE and DCR in this work to illustrate how prior work can be utilized in combination with $\{1\}$ -ADE to achieve state-of-the-art results; however, there may be a bespoke protocol from DDH or other, more common assumptions that achieve similar results. In particular, we felt that the works of [CNs07, GH08, GH11, JL09, KNP10, KNP11, LLM⁺17, RKP09, Zha11] were exploring methods by which to achieve succinct OT from DDH already; however, these protocols are limited to k out of n simulation-based OT since such settings are the only ones for which succinct simulation-based protocols could exist. Now that the notion of ADOT exists and supports a notion of succinctness for any opening regime, we hope that future efforts can be applied to finding succinct ADOT schemes.

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Appendices

A Relational Theorems

Theorem 5 ($\mathcal{X}_n^{1/2}$ -ADE implies One-Time Semantically-Secure Symmetric Key Encryption). *If succinct standard model $\mathcal{X}_{2n}^{1/2}$ -ADE (Definition 11) exists, then there exists a symmetric key encryption scheme with succinct keys and one-time semantic security.*

Proof.

Construction. First decide on some method for choosing n from $m = |\overline{m}|$ and λ such that

$$T_{ADE}(\lambda, m/(2n), 2n) = o(m)$$

Some such choice exists if the ADE scheme is succinct. Then to encrypt $\overline{m}$, for each $m/(2n)$ -length portion, make an additive share of the message portion to end up with $\mathbf{m} = \{[\overline{m}_i]_0, [\overline{m}_i]_1\}_{i \in [m/(2n)]}$. Then encrypt $\mathbf{m}$ using the ADE scheme and let the key be two openings (decided in advance) that jointly open the whole message. To decrypt, simply open the ciphertext twice and then sum each of the shares.

Correctness and Succinctness. Correctness follows immediately from ADE correctness and the correctness of the shares. Note that the openings are asymptotically smaller than the encrypted message, and so sending 2 openings is still asymptotically smaller.

Security. Consider the security of the scheme for two messages $\overline{m}_0$ and $\overline{m}_1$. We can rephrase the encryption portion of the protocol as sampling $\mathbf{m}$ from a distribution D_b that is dependent on $\overline{m}_b$ that creates the block shares and encrypts that. We point out that D_0 and D_1 are $\mathcal{X}_{2n}^{1/2}$ -indistinguishable distributions. As such, even if the adversary were to receive any $\mathcal{X}_{2n}^{1/2}$ opening of their choice, which distribution was output would be hidden, which by extension also hides the encrypted message. As such, the adversary cannot distinguish the ciphertexts given no openings whatsoever. $\square$

Theorem 6 (Succinct $\mathcal{X}_n$ -ADE implies $\{1\}$ -ADE). *Let Π be an $\mathcal{X}_n$ -ADE scheme (Definition 11) such that for noticeable n there exist $x_n \in \mathcal{X}_n$ such that $T_{ADE}(\lambda, m, n) = o(mHW(x_n))$ for Hamming weight $HW(x_n)$ of x_n .²⁵ Then Π can be used directly as an $\{1\}$ -ADE scheme (Definition 10).*

Proof. We point out that the message of interest $\overline{m}$ can be encoded into $\mathbf{m}_{x_n}$ directly, with all the other portions of $\mathbf{m}$ set to 0. Then the key to open the encryption is just $\mathbf{k}_{x_n}$, which has size $o(|\overline{m}|) = o(mHW(x_n))$ by definition. $\square$

Construction 5 ($\mathcal{X}_n$ -ADE from $\mathcal{X}_n$ -ADOT). *Let the given $\mathcal{X}_n$ -ADOT scheme require that the receiver online phase randomness (i.e. R_1) have length bounded by $r(\lambda, m, n)$. Let $\text{Dummy}_L(\tau)$ be an interactive procedure that simply responds as the left-hand-party in some interactive protocol according to transcript τ and then outputs $\perp$.*

$Enc(1^\lambda, \mathbf{m})$	
1 :	$R_0 \leftarrow_{\$} \{0, 1\}^\infty$
2 :	$R_1 \leftarrow_{\$} \{0, 1\}^{r(\lambda, m, n)}$
3 :	$s_0, r_0, \tau_0 \leftarrow \left\langle \text{Send}_0(1^\lambda, \mathbf{m}), \text{Recv}_0(1^\lambda, 1^m, 1^n; R_0) \right\rangle$
4 :	return $((R_0, \tau_0), (s_0, r_0, R_1))$

²⁵In other words, we count the number of 1s in the string x_n , which corresponds to the number of openings.

$Open(k, x)$
$1: (s_0, r_0, R_1) \leftarrow k$
$2: -, -, \tau_1 \leftarrow \langle Send_1(s_0), Recv_1(r_0, x; R_1) \rangle$
$3: \textbf{return } (R_1, \tau_1)$

$Dec(k_x, x, c)$
$1: (R_0, \tau_0) \leftarrow c$
$2: (R_1, \tau_1) \leftarrow k_x$
$3: -, r_0, - \leftarrow \langle Dummy_L(\tau_0), Recv_0(1^\lambda, 1^m, 1^n; R_0) \rangle$
$4: -, \mathbf{m}', - \leftarrow \langle Dummy_L(\tau_1), Recv_1(r_0, x; R_1) \rangle$
$5: \textbf{return } \mathbf{m}'$

Theorem 7 ($\mathcal{X}_n$ -ADE from $\mathcal{X}_n$ -ADOT). *$\mathcal{X}_n$ -ADE (Definition 11) can be generally constructed from $\mathcal{X}_n$ -ADOT (Definition 12) with online-cost $T(\lambda, m, n) \leq T_{ADOT}(\lambda, m, n) + r(\lambda, m, n)$.*

Proof.

Correctness. Because we only care about security for the sender, it is fine that the sender selects the randomness for the receiver in order to be able to leverage the ADOT scheme to quickly compute a decryption key τ_1 .

Also note that the transcript τ_1 is bounded in size by the online-phase size of the ADOT scheme. If we only send the transcript consisting of sender messages then it could potentially be smaller, which accounts for the inequality in the theorem statement.

Security. The adversary's view in the ADE scheme is the same as the receiver's in the ADOT scheme. Thus, ADE security (Definition 11) of the above scheme holds immediately, since adaptive distributional sender privacy (Definition 12) ensures distributional indistinguishability as well. $\square$

Theorem 8 (Universal ADE from ADGC). *Universal ADE (Definition 11) can be generally constructed from ADGC (Definition 13) with online-cost $T(\lambda, m, n) = T_{ADGC}(\lambda, C')$ for circuit C' that checks $x \in \mathcal{X}_n$ and encodes the output messages projectively from the input.*

Proof. We first give a very short construction of ADE that we will show the theorem for.

Construction. Given messages $\mathbf{m}$, the sender devises a circuit $C'_{\mathbf{m}}(x)$ that simply checks that $x \in \mathcal{X}_n$ and outputs $\mathbf{m}_x$ if in the set, and $\mathbf{m}_{0^n}$ otherwise. Then the sender simply follows the garbling procedure and also acts as the challenger—being send x in plaintext and responding with labels directly.

Correctness. Correctness follows directly from the garbling procedure.

Security. The garbling scheme maintains adaptive distributional privacy (Definition 13), and so the output $\mathbf{m}_x$ is also private just as in the ADE security game (Definition 11). $\square$

Theorem 9 (Universal ADOT from ADGC and $\mathcal{X}_n^{1/2}$ -ADOT). *Universal ADOT (Definition 12) for any opening ensemble $\mathcal{X}_n$ can be generally constructed from a projective ADGC and $\mathcal{X}_{2n}^{1/2}$ -ADOT (Definitions 12 and 13) with online-cost $T(\lambda, m, n) = T_{ADOT}(\lambda, \lceil |\tilde{x}|/n \rceil, 2n) + T_{ADGC}(\lambda, C')$ where C' is the circuit that computes $C'(x) = \mathbf{m}_x$.*

Proof. This construction looks similar to the construction used in the above proof of Theorem 8, except that we will be using $\mathcal{X}_{2n}^{1/2}$ -ADOT to make the receiver's input private.

Construction. Given messages $\mathbf{m}$, the sender devises a circuit $C'_{\mathbf{m}}(x)$ that simply outputs $\mathbf{m}_x$. Then the sender simply follows the ADGC garbling procedure and uses ADOT to transfer the input labels to the receiver obliviously.

Correctness. Correctness follows immediately from the ADOT scheme for the labels over the ADGC scheme, and likewise for the ADGC assuming the labels were correctly handed over.

Adaptive Distributional Sender Privacy. Let D_0 and D_1 be $\mathcal{X}_n$ -indistinguishable distributions that are the targets of the new $\mathcal{X}_n$ -ADOT scheme.

Consider new distributions D'_0 and D'_1 defined below that mimic the above protocol but with an idealized $\mathcal{X}_{2n}^{1/2}$ -ADOT functionality. Recall that we required the ADGC scheme to be projective, and so e should be of the form of $2n$ distinct labels.

$D'_b(\lambda, m, n)$	
1 :	$\sigma, \xi, \mathbf{m} \leftarrow \$ D_b(\lambda, m, n)$
2 :	$\tilde{C}, e, d \leftarrow \text{Garble}(1^\lambda, C'_{\mathbf{m}})$
3 :	$\xi'(x, \eta) := (\xi(x, \eta), d)$
4 :	return $((\sigma, \tilde{C}), \xi', e)$

We note that distributions D'_0 and D'_1 are $\mathcal{X}_{2n}^{1/2}$ -indistinguishable by ADGC adaptive distributional privacy. As such, the $\mathcal{X}_{2n}^{1/2}$ -ADOT maintains adaptive distributional sender privacy.

Also note that because we only reapply distributional security recursively a constant number of times, the multiplicative factor loss is also a constant and so we retain only a polynomial loss in security instead of an exponential loss.

Adaptive Simulation Receiver Privacy. Recall that ADGC is non-interactive except for the use of $\mathcal{X}_{2n}^{1/2}$ -ADOT. As such, adaptive simulation receiver privacy reduces to the same privacy guarantee from $\mathcal{X}_{2n}^{1/2}$ -ADOT. □

Theorem 10 (ADGC from $\{1\}$ -ADE and Adaptively Secure Garbled Circuits). *Construction 3 is an ADGC scheme (Definition 13) from $\{1\}$ -ADE and an adaptively secure garbling scheme (Definitions 5 and 10) with online-cost $T(\lambda, C) = T_{ADE}(\lambda, |d|, 1)$ for decoding table d .*

Proof.

Construction. The protocol simply encrypts the decoding table d , sending the ciphertext in the offline-phase. Then the garbler sends the decryption key in the online-phase as the new decryption table.

Correctness. The construction simply applies encryption d , which is then revealed on schedule, and so the evaluator has the proper material for evaluating the circuit.

Adaptive Distributional Privacy. Let D_0 and D_1 be indistinguishable circuit distributions. Then consider the following statement about the games from Definition 5 for $\sigma_b, \xi_b, C_b \leftarrow \$ D_b$ for $b \in \{0, 1\}$ and efficient adversary $\mathcal{A}$ that only queries $\xi_b(x_b, \cdot)$ once after the offline-phase is past.

$$\begin{aligned} \text{Real}_{\Pi, \mathcal{A}^{\sigma_0, \xi_0, C_0}}^{\text{sim-prv}}(\lambda) &\approx \text{Ideal}_{\Pi, \mathcal{A}^{\sigma_0, \xi_0, S, C_0}}^{\text{sim-prv}}(\lambda) \\ &\approx \text{Ideal}_{\Pi, \mathcal{A}^{\sigma_1, \xi_1, S, C_1}}^{\text{sim-prv}}(\lambda) \approx \text{Real}_{\Pi, \mathcal{A}^{\sigma_1, \xi_1, C_1}}^{\text{sim-prv}}(\lambda) \end{aligned}$$

The jump from Real to Ideal holds due to adaptive simulation privacy, and then jump between the Ideal games is due to the circuit indistinguishability of the distributions D_0 and D_1 .

We can then consider viewing the above garbling scheme as a distribution on which $\{1\}$ -ADE can apply. Consider distributions D'_0 and D'_1 implemented in the following way.

$D'_b(\lambda, m' = d , n' = 1)$	
1 :	$\sigma, \xi, C \leftarrow \$ D_b(\lambda, m, n)$
2 :	$\tilde{C}, e, d \leftarrow \text{Garble}(1^\lambda, C)$
3 :	$\xi'(1, (x, \eta)) := (\xi(1, (x, \eta)), \text{Encode}(e, x))$
4 :	$\bar{m} \leftarrow d$
5 :	return $(\sigma, \xi', \bar{m})$

Then we see that by the real game indistinguishability we showed earlier that the distributions D'_0 and D'_1 are $\mathcal{X}'_n$ -indistinguishable for $\mathcal{X}'_1 = \{1\}$. Then we see that we can immediately apply $\{1\}$ -ADE to get an adaptively secure ADGC scheme by encrypting the only element d . $\square$

B Multiplier-1 ADE

We formally introduce the notion of multiplier-1 ADE (M1-ADE), which is a strengthening of ADE that is designed to allow for non-modifying drop-in use to get adaptive security for existing protocols.

Definition 14 (Multiplier-1 $\mathcal{X}_n$ -ADE). *An $\mathcal{X}_n$ -ADE scheme Π is **multiplier-1** ($\mathcal{X}_n$ -M1-ADE) if for any two $\mathcal{X}_n$ -indistinguishable distributions D_0 and D_1 with advantage μ_D , the advantage of Π encrypting D_0 and D_1 is*

$$\mu_\Pi(\lambda, m, n) = \mu_D(\lambda, m, n) + \mu(\lambda, m, n, |\sigma|, |\xi|)$$

for negligible μ , which is independent of μ_D . Note that $|\sigma|$ and $|\xi|$ denote the size bound for the respective elements output by either D_0 or D_1 , w.l.o.g. it can be assumed that these elements always have the maximum size by padding. Also note that the adversarial advantage given by μ_Π, μ_D, μ goes up as $m, n, |\sigma|, |\xi|$ increase, but notably is a negligible function when $m, n, |\sigma|, |\xi|$ are any polynomial in λ .

We also show that the encryption scheme of Construction 1 is a universal M1-ADE scheme. The proof is nearly the same as the one for Theorem 1, but we are more careful about computing the adversarial advantage.

Theorem 11 (Universal M1-ADE from ROM). *The simple ROM encryption scheme (Construction 1) is also an $\mathcal{X}_n$ -M1-ADE scheme for any opening ensemble $\mathcal{X}_n$ (Definition 11) with online-cost $T(\lambda, m, n) = n\lambda$*

Proof. We first note that Construction 1 is clearly correct for the opening function of simply sending the keys associated with openings, and has size λ keys per encryption, and so the online-cost is $T(\lambda, m, n) = n\lambda$.

Security. To show security we introduce the two following games. Note that we will assume that the distinguisher is given the oracle $\mathcal{O}$ at the end of the protocol. We also give the distribution D and output function ξ oracle access so that the scheme is oracle-composable and so that the scheme is closer to a standard model game, since D and ξ would always be able to depend on any standard model implementation.

$\text{Game}_{\Pi, \mathcal{A}, D}^{\text{ade}}(\lambda, m, n)$	$\text{Hyb}_{\Pi, \mathcal{A}, D}(\lambda, m, n)$
1 : $\sigma, \xi, \mathbf{m} \leftarrow \$ D^{\mathcal{O}}(\lambda, m, n)$	1 : $\{\mathbf{k}_i \leftarrow \$ \{0, 1\}^\lambda\}_{i \in [n]}$
2 : $\{\mathbf{k}_i \leftarrow \$ \{0, 1\}^\lambda\}_{i \in [n]}$	2 : $\{c_i \leftarrow \$ \{0, 1\}^m\}_{i \in [n]}$
3 : $\{c_i \leftarrow \mathcal{O}(\mathbf{k}_i) \oplus \mathbf{m}_i\}_{i \in [n]}$	3 : $\sigma, \xi, \mathbf{m} \leftarrow \$ D^{\mathcal{O}}(\lambda, m, n)$
4 : $x, \eta, S \leftarrow \mathcal{A}^{\mathcal{O}}(1^\lambda, 1^m, 1^n, \sigma, \{c_i\}_{i \in [n]})$	4 : $x, \eta, S \leftarrow \mathcal{A}^{\mathcal{O}}(1^\lambda, 1^m, 1^n, \sigma, \{c_i\}_{i \in [n]})$
5 : return $(S, \xi^{\mathcal{O}}(x, \eta), \{\mathbf{k}_i\}_{x_i=1})$	5 : $z \leftarrow \xi^{\mathcal{O}}(x, \eta)$
	6 : program $\{\mathcal{O}(\mathbf{k}_i) := c_i \oplus \mathbf{m}_i\}_{x_i=1}$
	7 : return $(S, z, \{\mathbf{k}_i\}_{x_i=1})$

Then we are tasked with showing for any $\mathcal{X}_n$ -indistinguishable distributions D_0 and D_1 with advantage bounded by μ_D that

$$\{\text{Game}_{\Pi, \mathcal{A}, D_0}^{\text{ade}}(\lambda, m, n)\} \approx_{\mu_\Pi} \{\text{Game}_{\Pi, \mathcal{A}, D_1}^{\text{ade}}(\lambda, m, n)\}$$

for

$$\mu_\Pi(\lambda, m, n, |\sigma|, |\xi|) = \mu_D(\lambda, m, n) + 2\mu(\lambda, m, n, |\sigma|, |\xi|)$$

where μ is independent of μ_D . We will do in two steps.

First we show that for $b \in \{0, 1\}$,

$$\{\text{Game}_{\Pi, \mathcal{A}, D_b}^{\text{ade}}(\lambda, m, n)\} \approx_\mu \{\text{Hyb}_{\Pi, \mathcal{A}, D_b}(\lambda, m, n)\}$$

for negligible μ that only scales with $\lambda, m, n, |\sigma|$, and output size of ξ_b , but is independent of μ_D . For the purposes of reaching a contradiction, say that the above games are distinguishable.

First consider only computationally bounded distributions D (that run with time some polynomial in λ, m, n).

We then consider two avenues of attack and show that neither is feasible.

- If we consider what is available to the adversary by line 4, we note that the distributions are only different due to the ciphertexts being drawn differently. In Game the ciphertexts are drawn from the image of the RO while in Hyb the ciphertexts are drawn from uniform. Given only Q many queries to the RO, it is not possible to distinguish between the two distributions.²⁶ In particular, note that the advantage of this attack does not scale at all with μ_D .
- Excepting the ciphertexts, the adversary can witness the RO being programmed at some point $\mathbf{k}_i$. Of course, this would require that any distinguishing D and $\mathcal{A}$ pair would need to query at least one $\mathbf{k}_i$ before line 6 — in other words, D , $\mathcal{A}$, or ξ would need to query a point $\mathbf{k}_i$ before line 6. Because the keys are drawn uniformly, the probability of this occurring is negligible. Also note that this issue is separate from the prior point since the RO inputs and output images are independent. Note again that the negligible function bounding this attack only scales with the probability of guessing a string from a uniformly chosen set given some number of queries the adversary and ξ make²⁷, and does not scale at all with the distinguishing probability bound μ_D .

Note, however, that the distribution D need not be efficient. Because D is not given the keys $\mathbf{k}_i$ or ciphertexts c_i , we can model D as part of the adversary that gets to potentially witness the entirety of the oracle before the game starts, producing a polynomially bounded amount of advice about the RO. This setting is known as the auxiliary-input ROM of [Unr07], and it was shown in [CDGS18, Unr07] to be equivalent within polynomial security loss to a setting in which the adversary gets to program polynomially

²⁶The distributions D_0 or D_1 can also make queries, but the effective queries of D_0 and D_1 can be bound by $|\sigma|$ and $|\xi|$ instead, as will be shown later in the proof.

²⁷Note that it is true that D_0 and D_1 can still make queries here, but this case will be dealt with later

many locations of the RO prior to the game starting. Because the oracle queries that the challenger makes are uniform, it is only with negligible probability that they will intersect with the adversarially programmed RO locations. This is to say that even though the distributions D_0 and D_1 are unbounded and can make unbounded numbers of queries to the RO, this can be simulated by knowing only polynomially many locations of the RO, scaling with $|\sigma|$ and $|\xi|$ and specifically not with μ_D .

Then the second step is to show

$$\{\text{Hyb}_{\Pi, \mathcal{A}, D_0}(\lambda, m, n)\} \approx \{\text{Hyb}_{\Pi, \mathcal{A}, D_1}(\lambda, m, n)\}$$

In the Hyb games, the adversary does not get any access to $\mathbf{m}$ before they make their decision x, η , and then the distinguisher only sees $\mathbf{m}_x$ and $\xi^{\mathcal{O}}(x, \eta)$ after, where the unopened part of the messages are simply not encoded anywhere. The information that the adversary has access to information theoretically mimics that of the $\mathcal{X}_n$ -indistinguishability games, otherwise the games are identical. As such, the indistinguishability of the Hyb games reduces to that of the $\mathcal{X}_n$ -indistinguishability games for D_0 and D_1 , with the same advantage μ_D .

As such, we find that the total adversarial advantage is

$$\mu_{\Pi}(\lambda, m, n, |\sigma|, |\xi|) = \mu_D(\lambda, m, n) + 2\mu(\lambda, m, n, |\sigma|, |\xi|)$$

and so the construction is multiplier-1. $\square$

B.1 Applications of M1-ADE

The main advantage of M1-ADE versus general ADE is that M1-ADE can be applied recursively polynomial numbers of times while ADE loses security potentially exponentially quickly when applied recursively.

To demonstrate, we will consider constructions of $\mathcal{X}_n^{1/2}$ -ADE.

From PRGs. A well studied area considers how standard encryption notions, such as semantic encryption or CPA/CCA encryption schemes, fail to be shown secure under the selective openings setting, in which the adversary sees encryptions first and then chooses which encryptions to open. The setting is hard to even define security for, hence variations such as non-committing encryption [Nie02], selective-openings security (SIM and IND) [BHY09, DNRS99, FHKW10, LDL⁺14, ORV14], etc. For the case of the naïve construction from PRGs, in which to encrypt each message Alice samples a fresh seed and computes $G(s_i) \oplus m_i$, there is no known concrete demonstration of “weird” behavior that the adversary can apply in the selective openings setting; however, selective opening security cannot be achieved directly because a simulator, which chooses which instances of G to replace with a random oracle call to form a reduction, cannot know in advance how the adversary might open the messages, except with at most log-many messages.

Another way to view this is that encryption another message with a succinct key for the selective openings setting amplifies the advantage of the adversary – somehow – in the number of messages that could be opened.

From ADE. The definition of ADE presented in this work towards solving this problem; however, ADE is not arbitrarily applicable to the selective openings problem. For example, consider attempting to construct $\mathcal{X}_n^{1/2}$ -ADE from $\mathcal{X}_2^{1/2}$ -ADE in the naïve way by encrypting every two messages with (say projective) $\mathcal{X}_2^{1/2}$ -ADE.²⁸ We might wish to show security by recursive hybrids, in which for some distribution that outputs messages $M \in \{0, 1\}^{n \times m}$, we can consider distributions D^i that are over $M \in \{0, 1\}^{2 \times m}$ in which the first $2(i - 1)$ messages are encrypted in pairs using $\mathcal{X}_2^{1/2}$ and the latter messages have yet to be encrypted.

Such recursive application of $\mathcal{X}_2^{1/2}$ -ADE does not generally work, however. Say that in the $\mathcal{X}_n^{1/2}$ -ADE game, the adversary can distinguish with advantage μ_D , and that applying $\mathcal{X}_2^{1/2}$ -ADE increases this advantage to $2\mu_D + \mu$, for some negligible μ . Then we see that the worst-case advantage of the adversary after the i -th re-application of $\mathcal{X}_2^{1/2}$ -ADE may be lower-bounded by $2^i \mu_D(\lambda, m, n)$, which is not negligible.

²⁸Recall that we provide a construction from $\{1\}$ -ADE (Theorem 8), which is implied by $\mathcal{X}_2^{1/2}$ -ADE, but requires a more thoughtful approach.

$$\begin{array}{lcl}
D^0 & | & \sigma' := \sigma \quad | \quad m_1 \quad m_2 \quad | \quad m_3 \quad m_4 \quad | \quad \cdots \quad | \quad m_{n-1} \quad m_n \quad | \\
D^1 & | & \sigma' := \sigma || c_1 || c_2 \quad | \quad k_1 \quad k_2 \quad | \quad m_3 \quad m_4 \quad | \quad \cdots \quad | \quad m_{n-1} \quad m_n \quad | \\
& & \vdots \\
D^{n/2} & | & \sigma' := \sigma || c_1 || \cdots || c_n \quad | \quad k_1 \quad k_2 \quad | \quad k_3 \quad k_4 \quad | \quad \cdots \quad | \quad k_{n-1} \quad k_n \quad |
\end{array}$$

Figure 3: Starting with $\mathcal{X}_n^{1/2}$ distributions D_0 and D_1 that each output some side information σ and ξ with messages $M \in \{0, 1\}^{n \times m}$, we can consider using projective $\mathcal{X}_2^{1/2}$ -ADE to replace the messages (given in the online phase) with succinct keys, providing the ciphertexts as side information. The above shows this in steps, in which D^i is built by applying $\mathcal{X}_2^{1/2}$ -ADE to the $2i - 1$ and $2i$ -th messages.

This is to say that ADE may amplify the advantage of the adversary to distinguish the two distributions by some constant (or worse!), which makes it unsuitable for recursively applying more than a constant number of times, which is what was experienced when using PRGs in the naive way.

From M1-ADE. By definition, M1-ADE does not multiplicatively increase the distinguishing advantage of the original distributions as a consequence of applying the encryption scheme. As a result, we find that the advantage of the adversary in the hybrids presented in Figure 3 can be bounded above by $\mu_D + (n/2)\mu$, which is still negligible. As such, M1-ADE does truly work as an all-around adaptive encryption scheme in that it can compose essentially an arbitrary number of times.

M1-ADE and Adaptive Garbled Circuits. Indeed, $\mathcal{X}_n^{1/2}$ -ADE does exist from $\mathcal{X}_2^{1/2}$ -ADE²⁹; however, the method presented in Figure 3 can be directly applied to garbled circuits in a way that ADE cannot be generally applied.

In particular, consider the garbling scheme of [AIK11], which garbles AND/NAND gates information-theoretically and uses a fan-out gate, which uses cryptography to refresh keys, maintaining some standard key length. If we use $\mathcal{X}_2^{1/2}$ -ADE to directly serve as the PRF in the protocol, we run into a worse problem than that presented in Figure 3. Because the output of the fan-out gates serve as the keys of future encryptions, we find that the messages m_i could actually be keys from the encryption of messages m_j for $j < i$! This means that there is no well-formed single distribution on which to apply $\mathcal{X}_n^{1/2}$ -ADE at the same time for each gate in the circuit, nor can we go one-by-one since re-application of ADE increases adversary advantage exponentially.

Of course, $\mathcal{X}_2^{1/2}$ -M1-ADE immediately solves this problem. Because we can treat each pair of messages separately (while all the other messages are considered side information), we find that the fact that some of the future messages happen to be keys from previous applications of M1-ADE does not effect the well-formed-ness of the distribution only considering two messages at a time (i.e. some D^i). As such, a proof does go through, yielding an adaptive garbled circuit scheme from [AIK11] *without modification*, from a plausibly standard-model assumption.³⁰

Implications. We would like to stress again that we feel that the main contribution of this work is not necessarily a non-modifying lift to adaptivity of some previously natural garbling scheme. Instead, we consider this result to be an exploration of security guarantees that may be required of the ROM when used to implement the PRF to get adaptive security without modification, especially given the impossibility result in [KKPW21], which states that CPA encryption is insufficient to achieve adaptive security of Yao’s scheme without modification.

²⁹The proof goes through $\{1\}$ -ADE, and even though the proof is not projective – the ciphertext is a garbled circuit over all of the inputs (Theorem 8) – the keys are projective for $\mathcal{X}_n^{1/2}$.

³⁰In contrast to some idealized model, like the ROM.

Of particular consideration is the fact that M1-ADE is a non-standard encryption definition, even more so than ADE, in which the extra (non-)increase to distinguishing advantage is the specific target of the definition, despite being an assumption regarding computational security.³¹

B.2 Constructions from M1-ADE

Theorem 12 (M1-ADE Compressability). *If there exists a succinct projective $\mathcal{X}_n$ -M1-ADE scheme (Definition 14) with online-cost $T(\lambda, m, n) = nO(m^\delta)\text{poly}(\lambda)$ for any $\delta < 1$, then there exists a projective $\mathcal{X}_n$ -M1-ADE scheme with online-cost $T'(\lambda, m, n) = n\text{poly}(\lambda)$.*

Proof. The idea here is that we will recursively encrypt the messages until some size that scales only in $\text{poly}(\lambda)$.

Let Π be some projective $\mathcal{X}_n$ -M1-ADE protocol with online complexity $T(\lambda, m, n) \leq cnm^\delta p(\lambda) = nO(m^\delta)\text{poly}(\lambda)$. First note that this means that each encryption key is of size $m^\delta p(\lambda)$. Then we will consider for a given λ when $m^\delta p(\lambda) \leq m$:

$$\begin{aligned} cm^\delta p(\lambda) &\leq m \\ \implies cp(\lambda) &\leq m^{1-\delta} \\ \implies c'p(\lambda)^{\frac{1}{1-\delta}} &\leq m \end{aligned}$$

Let $p'(\lambda) = c'p(\lambda)^{\frac{1}{1-\delta}}$. What this tells us is that encrypting messages longer than $p'(\lambda)$ will result in a key that is shorter than the original message by at least one bit.

Protocol. Consider the following construction of Π' from $\Pi = (\text{Enc}, \text{Dec})$.

$\text{Enc}'(1^\lambda, \overline{m})$	$\text{Dec}'(k, c)$
1 : $c_1, k_1 \leftarrow \text{Enc}(\overline{m})$	1 : $(i, \{c_j\}_{j \in [i]}) \leftarrow c$
2 : $k \leftarrow k_1$	2 : $k_i \leftarrow k$
3 : $i \leftarrow 1$	3 : for $j \in [i - 1]$:
4 : while $ k > p'(\lambda)$:	4 : $k_{i-j} \leftarrow \text{Dec}(k_{i-j+1}, c_{i-j+1})$
5 : $c_{i+1}, k_{i+1} \leftarrow \text{Enc}(k_i)$	5 : $\overline{m} \leftarrow \text{Dec}(k_1, c_1)$
6 : $k \leftarrow k_{i+1}$	6 : return $\overline{m}$
7 : $i \leftarrow i + 1$	
8 : return $((i, \{c_j\}_{j \in [i]}), k)$	

Correctness. First note that the number of ciphertexts produced by Enc' will be bounded by m since each encryption decreases the key size by at least 1 bit as long as keys are longer than $p'(\lambda)$. As such, the encryption scheme produces a polynomial-sized telescoping encryption and decrypts this, where the one key output per encryption is at most size $p'(\lambda) = \text{poly}(\lambda)$.

Security. We will argue that Π' is secure by considering for each m and λ protocol Π'_i that only produces up to i telescoping encryptions (adding a cap in line 4) and showing that these are secure by induction.

For the base case, note that Π'_1 is the same as Π and so is M1-ADE by definition with some adversarial advantage bounded by

$$\mu_{\Pi_1}(\lambda, m, n, |\sigma|, |\xi|) = \mu_D(\lambda, m, n) + \mu(\lambda, m, n, |\sigma| + \lceil \log m \rceil, |\xi|)$$

³¹Such specific claims are of course more popular in the information-theoretic setting since one can be very precise about adversarial advantage.

The $\lceil \log m \rceil$ term from the counter provided with the ciphertext, which counts up only to m .

For the induction step, assume that $\Pi'_i = (\text{Enc}'_i, \text{Dec}'_i)$ is M1-ADE with advantage bounded by

$$\mu_{\Pi_i}(\lambda, m, n, |\sigma|, |\xi|) \leq \mu_D(\lambda, m, n) + i \cdot \mu(\lambda, m, n, |\sigma|) + (i-1) \cdot p_c(\lambda, m) + \lceil \log m \rceil, |\xi|)$$

where $p_c(\lambda, m)$ is the length of ciphertexts Π produces. Then consider $\mathcal{X}_n$ -indistinguishable distributions D_0 and D_1 for any opening ensemble $\mathcal{X}_n$. We construct the following new distributions.

$D'_b(\lambda, m, n)$	
1 :	$\sigma, \xi, \mathbf{m} \leftarrow \$ D_b(\lambda, m, n)$
2 :	$\{c_j, \mathbf{k}_j \leftarrow \text{Enc}'_i(\lambda, \mathbf{m}_j)\}_{j \in [n]}$
3 :	$((\sigma, \{c_j\}_{j \in [n]}), \xi, \{\mathbf{k}_j\}_{j \in [n]})$

Because Π'_i is $\mathcal{X}_n$ -M1-ADE, we have that D'_0 and D'_1 are also $\mathcal{X}_n$ -indistinguishable since the distinguishing probability state above is negligible in λ , given polynomial bounds in λ for $m, n, |\sigma|, |\xi|$. Thus, by encrypting the key outputs of D'_b with Π and appending the i -th ciphertext with c_i , we find that the new adversarial advantage is bounded by

$$\begin{aligned} & \mu_{\Pi_{i+1}}(\lambda, m, n, |\sigma|, |\xi|) \\ & \leq \mu_D(\lambda, m, n) + i \cdot \mu(\lambda, m, n, |\sigma|) + (i+1) \cdot p_c(\lambda, m) + \lceil \log m \rceil, |\xi|) \\ & \quad + \mu(\lambda, m, n, |\sigma| + i \cdot p_c(\lambda, m) + \lceil \log m \rceil, |\xi|) \\ & \leq \mu_D(\lambda, m, n) + i \cdot \mu(\lambda, m, n, |\sigma|) + i \cdot p_c(\lambda, m) + \lceil \log m \rceil, |\xi|) + \mu(\lambda, m, n, |\sigma| + i \cdot p_c(\lambda, m) + \lceil \log m \rceil, |\xi|) \\ & = \mu_D(\lambda, m, n) + (i+1) \cdot \mu(\lambda, m, n, |\sigma|) + i \cdot p_c(\lambda, m) + \lceil \log m \rceil, |\xi|) \end{aligned}$$

which is negligible.

First note that the number of recursive steps is bounded by m , since the length of the keys decreases by at least 1 bit each time. As such, we find that the adversarial advantage of Π_m is bounded above by

$$\mu_{\Pi_m}(\lambda, m, n, |\sigma|, |\xi|) \leq \mu_D(\lambda, m, n) + m \cdot \mu(\lambda, m, n, |\sigma|) + (m-1) \cdot p_c(\lambda, m) + \lceil \log m \rceil, |\xi|)$$

which gives only additive advantage on μ_D , and so every scheme Π_j for $j \leq m$ is M1-ADE.³² $\square$

[AIK11]/Yao's Style Construction. We find it simplest to explain how M1-ADE augments an existing garbled circuit scheme with provable adaptive scheme by leveraging the Boolean GC scheme of [AIK11]. Here, AND/NAND gates are handled via *information-theoretic* garbling techniques, where we garble the circuit from outputs to inputs. The main challenge—and where a cryptographic assumption is applied—is in the handling of wire fan-out. For this, [AIK11] proposed a *key shrinking gadget*, which, in the selective setting, can be instantiated from any PRG. We will simply replace this PRG by an appropriate use of M1-ADE, and the resulting scheme will be adaptively secure.

In more detail, to garble an AND gate over inputs x and y such that the output bit uses arbitrary ℓ -bit string labels L_0 and L_1 , where string multiplication multiplies per bit index-wise while bits scale strings. The garbling procedure will assign labels $A_1||A_2$ for input wire x and labels $A_3||A_4$ for input wire y .

³²Note that these schemes are meant to be used with respect to j (or m) which scales in some polynomial of λ , and the above advantage of course should still maintain a multiplier 1 correspondence even when m or j scales with λ .

GarbleAND(p, L_0, L_1)	
1 :	$B_1, B_2, B_4 \leftarrow_{\$} \{0, 1\}^\ell$
2 :	$B_3 \leftarrow B_1 B_4 + B_2 + L_p$
3 :	$\Delta \leftarrow L_0 + L_1$
4 :	$A_1^x \leftarrow B_1 + x\Delta$
5 :	$A_2^x \leftarrow B_2 + xB_4\Delta$
6 :	$A_3^y \leftarrow B_3 + yB_1$
7 :	$A_4^y \leftarrow B_4 + y\mathbf{1}$
8 :	return $(A_1^0 A_2^0, A_1^1 A_2^1, A_3^0 A_4^0, A_3^1 A_4^1)$

Then the labels are $A_1^x||A_2^x$ associated with x and $A_3^y||A_4^y$ associated with y , where p chooses between AND and NAND. To compute the AND we compute

$$L_{xy \oplus p} = A_1 A_4 + A_2 + A_3$$

The garbling scheme is information theoretic in the sense that given only $A_1^x||A_2^x$ and $A_3^y||A_4^y$ for some choice of x and y , the information about $L_{1-xy \oplus p}$ and p is information theoretically lost.

We will construct a garbling scheme for circuits C over AND/NAND gates and fan-out gates. The fan-out gates take in a wire and output any number of wires, where the output wires are set to be the same value as the input wires.

The following gates have the below meta information associated with them in the circuit.

- AND/NAND: (p, w_1, w_2, w_3) for parity p , input wires w_1, w_2 , and output wire w_3 .
- fan-out: (w_1, W) for input wire w_1 and output wire set W .

Then the construction follows, noting that the gates are topologically ordered.

Construction 6 (Garbled Circuits of [AIK11] from $\mathcal{X}_2^{1/2}$ -M1-ADE). *Let $\Pi = (Enc, Dec)$ be a projective $\mathcal{X}_2^{1/2}$ -M1-ADE scheme (Definition 14) such that $T_{ADE}(\lambda, m, n) = n\text{poly}(\lambda)$. Then consider the following construction of a garbling of fan-out gates given the output-wire labels.*

GarbleFO($1^\lambda, W$)	
1 :	$\{(L_{0,i}, L_{1,i})\}_{i \in [W]} \leftarrow W$
2 :	$p \leftarrow_{\$} \{0, 1\}$
3 :	$L_0 \leftarrow L_{0,1} \dots L_{0, W } p$
4 :	$L_1 \leftarrow L_{1,1} \dots L_{1, W } (1 - p)$
5 :	$c_0, k_0 \leftarrow Enc(1^\lambda, L_0)$
6 :	$c_1, k_1 \leftarrow Enc(1^\lambda, L_1)$
7 :	return $(c_p c_{1-p}, (k_0 p, k_1 (1 - p)))$

The garbling of fan-out gates proceeds in the natural manner implied by GarbleFO and point-and-permute³³ — i.e. in reverse topological order. Given GarbleAND and GarbleFO, a projective garbling scheme is induced where each gate is garbling **back to front** using output labels for the per-gate garbling procedures to get input labels. The garbled circuit consists of the garbled material output by GarbleFO (since GarbleAND does not produce garbled material), while encoding string e consists of the input-wire labels, and decoding string d consisting of the output wire labels.

³³Note that the outputs of AND/NAND gates cannot go directly into another AND/NAND gate without going into a fan-out gate in order to keep key sizes small

For simulation-based security, the decoding table is sent in the online-phase, while for ADS, the decoding table is encrypted using $\{1\}$ -ADE with the ciphertext sent in the offline-phase and the key sent in the online-phase.

Remark 6 (Inverting Fan-Out). *Note that a variant of this scheme might be considered in which the fan-out gates can also invert some of the output wires by swapping them in secret. We do not use that version in this paper to keep the construction simple, but the proof of security should look exactly the same.*

Now we show that Construction 6 is a ADGC scheme.

Theorem 13 (ADGC from M1-ADE). *Construction 6 is an AGC scheme (Definition 5) with online-cost $T(\lambda, C) = m\text{poly}(\lambda) + 2n\lambda$ or an ADGC scheme (Definition 13) from M1-ADE with online-cost $T(\lambda, C) = m\text{poly}(\lambda)$ if $T_{M1-ADE}(\lambda, m', n') = n'\text{poly}(\lambda)$.*

Proof. We first give a brief argument towards correctness.

Correctness. In the construction the induced garbling scheme should immediately be correct without considering the size of the garbled circuit. Naturally, attempting to use information theoretic garbling without compressing the labels would result in exponential blowup in the label size in the depth of the circuit; however, because our M1-ADE scheme only scales with λ and the output of AND/NAND gates must go through M1-ADE, we find that the maximum label size is 2 times the M1-ADE key size on any given wire.

Adaptive Distributional Obliviousness. The first part of the security proof will be to build up a distributionally oblivious garbling scheme with respect to some circuit C drawn from one of two circuit indistinguishable distributions. Let $\text{Garble}_i(1^\lambda, C)$ be a garbling procedure that acts on circuit C in the following way:

1. Sample λ -length output labels for each output wire with point-and-permute, the choices here are stored in decoding table d .
2. Garbling backwards in topological order, apply each appropriate gate as described in Construction 6 up to the i -th gate in C to get partially garbled circuit $\tilde{C}_i$.
3. For each wire that is broken, i.e. in C would come from an input or gate that has not been garbled yet, create a list of each input label in order of wire ordered by point and permute bit into $\mathbf{L}$.
4. output $(\tilde{C}_i, \mathbf{L}, d)$.

Note that $\Pi_{|C|}$ is the original garbling scheme presented in Construction 6, sans the decoding table sent in the offline-phase.

We also consider the following distributions., note that $x \in \{0, 1\}^m$

$D_b^i(\lambda, m := \text{poly}(\lambda), n := \mathbf{L})$	
1:	$\sigma, \xi, C \leftarrow \$ D_b(\lambda, m, n)$
2:	$(\tilde{C}_i, \mathbf{L}, d) \leftarrow \text{Garble}_i(1^\lambda, C)$
3:	$\sigma_i \leftarrow (\sigma, \tilde{C}_i)$
4:	$\xi_i(x', (x, \eta)) := \xi(x, \eta)$
5:	return $(\sigma^i, \xi^i, \mathbf{L})$

We then proceed by induction. For the base case, we find that D_0^0 and D_1^0 are $\mathcal{X}_{2^n}^{1/2}$ -indistinguishable distributions since the garble procedure just outputs the uniform output labels, which the adversary would then choose to open, along with seeing side information related to the circuit C draw from the circuit

indistinguishable distributions. We note that the distinguishing advantage is then upper bounded by the distinguishing rate of the original circuit distributions, which we will call μ_C .

For the induction step, assume that D_0^i and D_1^i have a distinguishing advantage bounded above by

$$\mu_i(\lambda, m, n, |\sigma|, |\xi|) \leq \mu_C(\lambda, m, n) + i \cdot \mu(\lambda, m, n, |\sigma| + i \cdot p_{FO}(\lambda), |\xi|)$$

where $p_{FO}(\lambda)$ is the size of the garbled material associated with one fan-out gate. Then we will show that D_0^{i+1} and D_1^{i+1} have distinguishing advantage upper bounded by the above formula for $i+1$ instead of i .

Case 1: AND/NAND. The first case is that the $i+1$ -th gate is an AND/NAND gate. Let w_1 and w_2 be the wires that feed into the AND gate and w_3 be the output wire – recall that the wires must be unique. Note that Garble_{i+1} will take the labels in $\mathbf{L}$ associated with wire w_3 and produces labels associated with wires w_1 and w_2 , putting the new labels into the new $\mathbf{L}'$. Note that when the adversary in D_b^{i+1} chooses an opening $L'_{x''}$, because the AND gate is information theoretically secure, this can be interpreted as a circuitous way to select an opening $L_{x'}$. As such, we find that the distinguishing advantage is unchanged, which is certainly less than the target rate for the induction step.

Case 2: Fan-Out. The second case is that the $i+1$ -th gate is a fan-out gate. Let w_1 be the input wire and W be the set of output wires – again recall that all of these wires must be unique. Then we find that Garble_{i+1} will take all the labels in $\mathbf{L}$ associated with W and coalesces them into one wire w_1 , where the labels associated with w_1 replace those in $\mathbf{L}'$ associated with W . Again, this can be seen as a weird way for the adversary to select between the wires in W , which also limits their options.

Of course, applying M1-ADE to D_b^i will increase the distinguishing advantage by

$$\mu(\lambda, m, n, |\sigma| + (i+1) \cdot p_{FO}(\lambda), |\xi|)$$

which results in advantage bounded by

$$\begin{aligned} & \mu_{i+1}(\lambda, m, n, |\sigma|, |\xi|) \\ & \leq \mu_C(\lambda, m, n) + i \cdot \mu(\lambda, m, n, |\sigma| + i \cdot p_{FO}(\lambda), |\xi|) + \mu(\lambda, m, n, |\sigma| + (i+1) \cdot p_{FO}(\lambda), |\xi|) \\ & \leq \mu_C(\lambda, m, n) + i \cdot \mu(\lambda, m, n, |\sigma| + (i+1) \cdot p_{FO}(\lambda), |\xi|) + \mu(\lambda, m, n, |\sigma| + (i+1) \cdot p_{FO}(\lambda), |\xi|) \\ & = \mu_C(\lambda, m, n) + (i+1) \cdot \mu(\lambda, m, n, |\sigma| + (i+1) \cdot p_{FO}(\lambda), |\xi|) \end{aligned}$$

Then after the induction, we have that $D_0^{|C|}$ and $D_1^{|C|}$ have distinguishing advantage bounded by

$$\mu_{|C|}(\lambda, m, n, |\sigma|, |\xi|) \leq \mu_C(\lambda, m, n) + |C| \cdot \mu(\lambda, m, n, |\sigma| + |C| \cdot p_{FO}(\lambda), |\xi|)$$

Which is negligible even for circuits with size bounded by any polynomial in λ . As such, we find that $\text{Garble}_{|C|} = \text{Garble}$ is “oblivious”, which we mean here that the above distributions are indistinguishable. We then use this to get ADGC.

AGC Privacy. We can reinterpret distributions $D_0^{|C|}$ and $D_1^{|C|}$ as being circuit indistinguishable for circuits $C'_b(x) := C_b(x)$ selects d for where the output of the circuit outputs labels from the decoding table d , for circuits C_b output from D_b . Then the proof follows similarly to Theorem 15. We select D_0 to output the fixed circuit C and side information $\xi_0(x, \eta) := d$, while D_1 outputs C' with only AND gates with side information $\xi_1(x, \eta) := d_{C(x)}$, where $d_{C(x)}$ takes the proper labels from the decoding table d associated with the outputs of $C(x)$, and replaces the other labels with uniform strings (still consistent with point-and-permute). D_0 and D_1 should be perfectly circuit indistinguishable. We find then that $D_0^{|C|}$ and $D_1^{|C|}$ should be indistinguishable, which implies that the garbling protocol maintains the indistinguishability. As such, we can design a simulator for the garbling scheme that simply samples from $D_1^{|C|}$ in the offline-phase and then outputs $\xi_1(x, \eta)$ for the online-phase.

Full ADGC. Note that after obtaining indistinguishability between $D_0^{|C|}$ and $D_1^{|C|}$, we find that setting the side information to $\xi'_b(x, \eta) := (C(x), d, \xi_b(x, \eta))$ we can then instead send the decoding table in the online-phase.³⁴ Then we find that we can use $\{1\}$ -ADE to encrypt the labels, sending a succinct decryption key in the online-phase. $\square$

C Adaptive Simulation from Adaptive Distributional Security

One can recover the simulation-based adaptivity definitions from adaptive distributional security by adding back in the online-phase one-time pad. We formally show this for each related security definition in the following.

Theorem 14 (Adaptive $\mathcal{X}_n$ -OT from $\mathcal{X}_n$ -ADOT). *Adaptive $\mathcal{X}_n$ -OT (Definition 2) can be generally constructed from $\mathcal{X}_n$ -ADOT (Definition 12) with online-cost $T(\lambda, m, n) = T_{ADOT}(\lambda, m, n) + mn$.*

Proof. We first give the transformation from ADOT.

Construction. Let $\mathbf{m}$ be the messages that are to be selected from by the receiver. The sender first uniformly samples $\mathbf{m}'$ from $\{0, 1\}^{m \times n}$ and then conducts ADOT over $\mathbf{m}'$, then sends $\mathbf{m} \oplus \mathbf{m}'$ in the online-phase afterwards, which the receiver uses to decrypt their messages.

Sender Privacy. We first consider viewing the above protocol as distributions. In the real distribution D_0 , there is no side information σ_0 , but we have that $\xi_0(x, \eta) := \mathbf{m}' \oplus \mathbf{m}$ with messages $\mathbf{m}'$. We compare this to a second distribution D_1 that also has no side information, and sets $\xi_0(x, \eta) := \mathbf{m}''$ with messages $\mathbf{m}_x$ such that $\mathbf{m}''_x = \mathbf{m}'_x \oplus \mathbf{m}_x$ while the other messages are uniform. We find that D_0 and D_1 are perfectly $\mathcal{X}_n$ -indistinguishable.

Since the above protocol does not depend at all on the encrypted messages until the one-time pad at the very end, and since ADOT is self-extracting, i.e. the adversary emits their choice x to the challenger automatically, a simulator can easily query the ideal functionality to obtain $\mathbf{m}_x$ and use the one-time pad in the online-phase to have the adversary decrypt $\mathbf{m}_x$, where the parts of the one-time pad corresponding to unopened messages is set to uniform.

Receiver Privacy. Simulation-based receiver privacy reduces directly to ADOT receiver privacy, because the sender receives the same messages from the receiver in both protocols. $\square$

Theorem 15 (Adaptive Garbled Circuits from ADGC). *Adaptive garbled circuits (Definition 5) can be generally constructed from ADGC (Definition 13) with online-cost $T(\lambda, C) = T_{ADGC}(\lambda, C') + m$ for C' having potentially $O(m)$ more gates³⁵.*

Proof. We first give the transformation from ADGC.

Construction. Let C be the desired circuit to show security for. Then to garble C first consider a new topology $\Phi(C')$ that can compute both $C(\cdot) \oplus y$ and C'_y for all constants $y \in \{0, 1\}^m$, where C' is a circuit that always outputs $y \in \{0, 1\}^n$. Then for uniform y compute $C(\cdot) \oplus y$ using ADGC with topology $\Phi(C')$ and provide y as the last message, which the evaluator can use to decipher the output. Note that $\Phi(C')$ should not add more than $O(m)$ gates, enough to xor the output conditioned on some constant.³⁶

³⁴Note that even though the output labels of the circuit are not decided by the circuit distributions, because the labels are perfectly indistinguishable between D_0^0 and D_1^0 with the only thing hidden being their permutation (which is encoded by $C_b(x)$), all the labels can be included for free.

³⁵ C' is constructed like C but outputs a uniform value that is known in advance, which usually does not require adding more gates for a garbling scheme that hides gates that output constants

³⁶In practice, most garbling schemes, and thus ADGC schemes, are able to hide which gate is being garbled, in which case the output gates can either be the true gates or const gates with the output value hardcoded.

Simulation Privacy. Let D_0 and D_1 be distributions that follow the template of indistinguishable circuit distributions (Definition 9) such that D_0 outputs $C(\cdot) \oplus y$ and $\xi(x) = y$ for uniformly sampled y and D_1 outputs C'_y and $\xi(x) = C(x) \oplus y$ for uniformly sampled y . We note that these two distributions are perfectly indistinguishable, and so we find that after garbling using ADGC that they should remain indistinguishable. Since the circuit of D_1 does not depend on x in the online-phase, a simulator can simply run the protocol using circuit C'_y , get the output from the adversary, and then set the decoding pad to $C(x) \oplus y$. $\square$

Theorem 16 (Adaptive 2PC from $\mathcal{X}_n^{1/2}$ -ADOT and Adaptive GC). *Adaptive 2PC (Definition 7) can be generally constructed from $\mathcal{X}_{2n}^{1/2}$ -ADOT and projective AGC (Definitions 5 and 12) with online-cost $T(\lambda, C) = T_{AGC}(\lambda, C) + T_{ADOT}(\lambda, m', 2n)$ where m' is the size of labels given by the AGC.*

Proof. We first provide the transformation.

Constructions. The construction is the same as the OT and GC construction for achieving MPC [BHR12b, Yao86], except with ADOT instead of OT. This is to say that OT is conducted over the input wire labels to the garbled circuit, since it should be projective. The labels corresponding to the garbler party's input can also be included in the ADOT, where the garbler party can just send the evaluator party a masked version of their input. For the security we will assume w.l.o.g. that the evaluator party is the only party with input due to the aforementioned masking technique.

Garbler Party Privacy. We start by reinterpreting the garbling privacy security games as indistinguishable circuit distributions (Definition 9). For convenience, we give the games from adaptive simulation privacy in Definition 5 here again.

$\text{Real}_{\Pi, \mathcal{A}, C}^{\text{sim-prv}}(\lambda)$	$\text{Ideal}_{\Pi, \mathcal{A}, \mathcal{S}, C}^{\text{sim-prv}}(\lambda)$
1 : $\tilde{C}_0, e_0, d_0 \leftarrow \text{Garble}(1^\lambda, C)$	1 : $\tilde{C}_1, \tilde{x}_1, S_S \leftarrow \mathcal{S}_1(1^\lambda, C)$
2 : $x, S \leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, \tilde{C})$	2 : $x, S_A \leftarrow \mathcal{A}(1^\lambda, 1^m, 1^n, \tilde{C})$
3 : $\tilde{x}_0 \leftarrow \text{Encode}(e_0, x)$	3 : $d_1 \leftarrow \mathcal{S}_2(S_S, C(x))$
4 : return $(S, \tilde{x}_0, d_0)$	4 : return $(S_A, \tilde{x}, d_1)$

Then consider distributions D_0 and D_1 such that D_b outputs $(\sigma_b, \mathbf{m}^b, \xi_b)$ that are sampled in the following way:

$$\begin{aligned}
\sigma_b &:= \tilde{C}_b \\
\mathbf{m}^0 &:= e_0 \\
\mathbf{m}^1 &:= \text{Interleave}(\tilde{x}_1) \\
\xi_b(x) &:= d_b
\end{aligned}$$

where $\text{Interleave}(\tilde{x})$ interleaves the values of $\tilde{x}$ so that no matter which choices the receiver makes they will always end up with $\tilde{x}$.

Then we see that by garbled circuit simulation privacy that distributions D_0 and D_1 must be $\mathcal{X}_{2n}^{1/2}$ -indistinguishable (Definition 8). As such, $\mathcal{X}_{2n}^{1/2}$ -ADOT can be used to implement the encoding function.

Evaluator Party Privacy. Note that the garbling step is non-interactive excepting the projective choice step. As such, we find that privacy of evaluator's input x comes down to receiver privacy from ADOT. $\square$

D Adaptive Simulation is Adaptive Distributional Security

We show in this section that adaptive simulation definitions are also adaptively distributional. Admittedly, the following theorems are very straightforward and similar; we include them for completeness.

Theorem 17 (Adaptive $\mathcal{X}_n$ -OT is $\mathcal{X}_n$ -ADOT). *Adaptive $\mathcal{X}_n$ -OT schemes (Definition 2) with sender privacy simulators that do not rewind back to the offline-phase once in the online-phase are also $\mathcal{X}_n$ -ADOT schemes (Definition 12).*

Proof.

Adaptive Distributional Sender Privacy. We first note that simulation security should be robust against side information σ and $\xi(x, \eta)$, since the adversary could know and run these objects themselves. For distributions D_0 and D_1 that are $\mathcal{X}_n$ -indistinguishable, we find that for $\sigma_b, \xi_b, \mathbf{m}^b \leftarrow \$ D_b$ the following holds.

$$\begin{aligned} \{\text{Real}_{\Pi, \mathcal{A}^{\sigma_0, \xi_0}, \mathbf{m}^0}^{\text{sim-ot}}(\lambda)\} &\approx \{\text{Ideal}_{\Pi, \mathcal{A}^{\sigma_0, \xi_0}, \mathcal{S}, \mathbf{m}^0}^{\text{sim-ot}}(\lambda)\} \\ &\approx \{\text{Ideal}_{\Pi, \mathcal{A}^{\sigma_1, \xi_1}, \mathcal{S}, \mathbf{m}^1}^{\text{sim-ot}}(\lambda)\} \approx \{\text{Real}_{\Pi, \mathcal{A}^{\sigma_1, \xi_1}, \mathbf{m}^1}^{\text{sim-ot}}(\lambda)\} \end{aligned}$$

for games taken from Definition 2, where the adversary is given σ_b and runs an oracle of ξ_b exactly once after the offline-phase on their input. We use the specific simulation fact here to make sure that the simulator cannot force the adversary to query ξ_b multiple times, since this is not allowed in the distinguishing games for $\mathcal{X}_n$ -indistinguishability. Note that the simulated worlds are indistinguishable due to the $\mathcal{X}_n$ -indistinguishability of the distributions.

Adaptive Simulation Receiver Privacy. This requirement is the same in both definitions, so it is trivially satisfied. $\square$

Theorem 18 (Adaptive Garbled Circuits is ADGC). *Adaptive garbled circuit schemes (Definition 5) are also ADGC schemes (Definition 13).*

Proof. The proof of this is nearly identical to that of Theorem 17. We write it here for completeness.

Adaptive Distributional Privacy. We note again that simulation security should be robust against side information σ and $\xi(x, \eta)$, since the adversary could know and run these objects themselves. For indistinguishable circuit distributions D_0 and D_1 , we find that for $\sigma_b, \xi_b, C_b \leftarrow \$ D_b$ that the following holds.

$$\begin{aligned} \{\text{Real}_{\Pi, \mathcal{A}^{\sigma_0, \xi_0}, C_0}^{\text{sim-prv}}(\lambda)\} &\approx \{\text{Ideal}_{\Pi, \mathcal{A}^{\sigma_0, \xi_0}, \mathcal{S}, C_0}^{\text{sim-prv}}(\lambda)\} \\ &\approx \{\text{Ideal}_{\Pi, \mathcal{A}^{\sigma_1, \xi_1}, \mathcal{S}, C_1}^{\text{sim-prv}}(\lambda)\} \approx \{\text{Real}_{\Pi, \mathcal{A}^{\sigma_1, \xi_1}, C_1}^{\text{sim-prv}}(\lambda)\} \end{aligned}$$

for games taken from Definition 5, where the adversary is given σ_b and runs an oracle of ξ_b exactly once after the offline phase on their input. Note that the simulated worlds are indistinguishable due to the circuit indistinguishability of the distributions. $\square$